

EasyNSShape - Shape Formula Reference / 形状参数方程手册

All 128 parametric shapes shipped in the `EasyNSShape` plugin. Every shape is one NDI dynamic-input script (the formula) plus one NE emitter and one NS system that consume it. EasyNSShape 插件全部 128 个参数化形状。每个形状由一个 NDI 动态输入脚本（参数方程）、一个消费它的 NE 发射器和一个 NS 系统组成。

2D shapes / 二维形状	89
3D shapes / 三维形状	27
3D special (multi-parameter) / 三维特殊形状	12
Total / 合计	128

Every NDI takes a single float `Module.InputT` (the curve parameter) and returns a `float2` Or `float3` position. The multi-parameter shapes add `InputU / InputV`, `InputA - InputC`, `InputR / InputRho / InputN`, `InputK`, `InputX0 - InputZ0` Or `InputP0 - InputP3`. 每个 NDI 接收一个浮点参数 `Module.InputT` (曲线参数)，返回 `float2` 或 `float3` 位置。

Each card lists the textbook math form first, then the HLSL that actually runs. Symbol mapping: 每张卡片先给出数学参数方程，再给出实际运行的 HLSL。符号对照：

Symbol	HLSL input
t	<code>Module.InputT</code> (curve parameter / 曲线参数)
u, v	<code>Module.InputU</code> , <code>Module.InputV</code>
a, b, c	<code>Module.InputA</code> , <code>Module.InputB</code> , <code>Module.InputC</code>
R, ρ, n, k	<code>Module.InputR</code> , <code>Module.InputRho</code> , <code>Module.InputN</code> , <code>Module.InputK</code>
x ₀ , y ₀ , z ₀	<code>Module.InputX0</code> , <code>Module.InputY0</code> , <code>Module.InputZ0</code>
P ₀ ... P ₃	<code>Module.InputP0 ... Module.InputP3</code> (control points / 控制点)

Icons: `Thumbs/<NS asset>.png`, 240x135. 图标： `Thumbs/<NS 资源名>.png` , 240x135。

Section / 章节	Jump / 跳转
2D Shapes / 二维形状 (89)	go
3D Shapes / 三维形状 (27)	go
3D Special (multi-parameter) / 三维特殊形状 (12)	go

2D Shapes / 二维形状 (89)

二维参数曲线 - 89

1. Archimedean Spiral - 阿基米德螺线



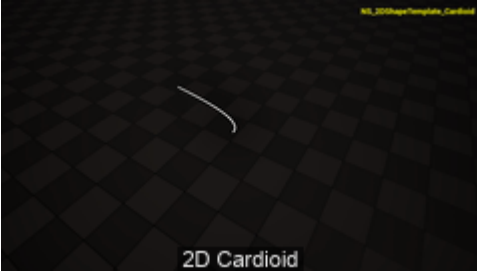
Field	Value
English	Archimedean Spiral
中文	阿基米德螺线
Formula (Math)	$x = t \cos t, y = t \sin t$
Formula (HLSL)	<code>OutVector2D = float2(InputT * cos(InputT), InputT * sin(InputT))</code>
Description	Archimedean spiral — radius grows linearly with the angle.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_ArchimedeanSpiral - NE NE_2DShapeTemplate_ArchimedeanSpiral - NS NS_2D_ArchimedeanSpiral - NFS NFS2D_ArchimedeanSpiral

2. Astroid - 星形线



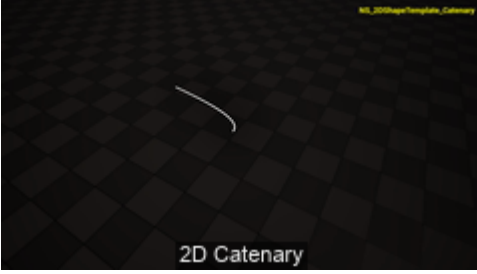
Field	Value
English	Astroid
中文	星形线
Formula (Math)	$x = \cos^3 t, y = \sin^3 t$
Formula (HLSL)	<code>float2(pow(cos(InputT), 3.0), pow(sin(InputT), 3.0))</code>
Description	Four-cusped hypocycloid (star curve).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Astroid - NE NE_2DShapeTemplate_Astroid - NS NS_2D_Astroid - NFS NFS2D_Astroid

3. Cardioid - 心形线



Field	Value
English	Cardioid
中文	心形线
Formula (Math)	$x = \cosh t, y = t$
Formula (HLSL)	<code>float2(cosh(InputT), InputT)</code>
Description	Cardioid (heart-shaped curve).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Cardioid - NE NE_2DShapeTemplate_Cardioid - NS NS_2D_Cardioid - NFS NFS2D_Cardioid

4. Catenary - 悬链线



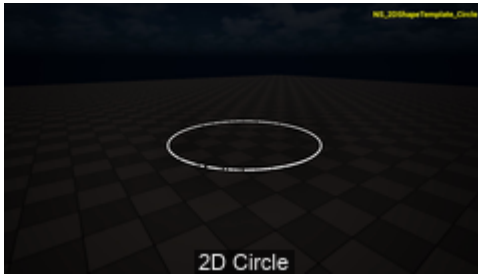
Field	Value
English	Catenary
中文	悬链线
Formula (Math)	$x = \cosh t, y = t$
Formula (HLSL)	<code>float2(cosh(InputT), InputT)</code>
Description	Catenary — the shape of a hanging chain (cosh).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Catenary - NE NE_2DShapeTemplate_Catenary - NS NS_2D_Catenary - NFS NFS2D_Catenary

5. Catenary 2 - 悬链线 2



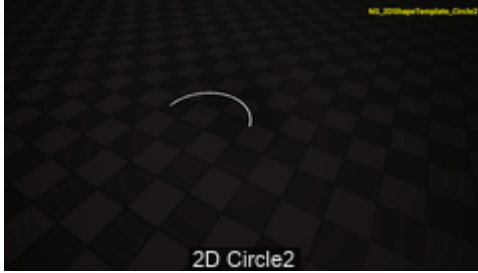
Field	Value
English	Catenary 2
中文	悬链线 2
Formula (Math)	$x = t, y = \cosh t$
Formula (HLSL)	<code>float2(InputT, cosh(InputT))</code>
Description	Catenary with the axes swapped.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Catenary2 - NE NE_2DShapeTemplate_Catenary2 - NS NS_2D_Catenary2 - NFS NFS2D_Catenary2

6. Circle - 圓



Field	Value
English	Circle
中文	圓
Formula (Math)	$x = \cos t, y = \sin t$
Formula (HLSL)	<code>float2(cos(InputT), sin(InputT))</code>
Description	Unit circle traced by cos/sin.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Circle - NE NE_2DShapeTemplate_Circle - NS NS_2D_Circle - NFS NFS2D_Circle

7. Circle 2 - 圓 2



Field	Value
English	Circle 2
中文	圓 2
Formula (Math)	$x = 1/(1 + t^2), y = t/(1 + t^2)$
Formula (HLSL)	<code>float2(1.0/(1.0+InputT*InputT), InputT/(1.0+InputT*InputT))</code>
Description	Rational-parameterised circle.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Circle2 - NE NE_2DShapeTemplate_Circle2 - NS NS_2D_Circle2 - NFS NFS2D_Circle2

8. Circle Center - 圓心圓



Field	Value
English	Circle Center
中文	圓心圓
Formula (Math)	$x = 2t^2/(1 + t^2), y = 2t/(1 + t^2)$
Formula (HLSL)	<code>float2(2.0*InputT*InputT/(1.0+InputT*InputT), 2.0*InputT/(1.0+InputT*InputT))</code>
Description	Circle whose parameterisation is centred on the origin.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CircleCenter - NE NE_2DShapeTemplate_CircleCenter - NS NS_2D_CircleCenter - NFS NFS2D_CircleCenter

9. Circle Nonlinear - 非线性圓



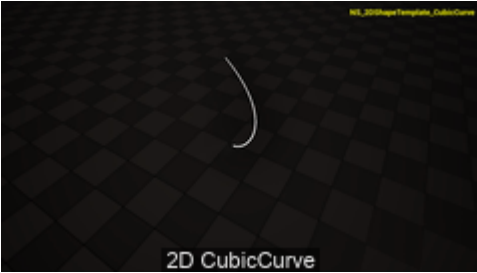
Field	Value
English	Circle Nonlinear
中文	非线性圓
Formula (Math)	$x = \sin(t^2), y = \cos(t^2)$
Formula (HLSL)	<code>float2(sin(InputT*InputT), cos(InputT*InputT))</code>
Description	Circle with a non-linear (quadratic) angle sweep.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CircleNonlinear - NE NE_2DShapeTemplate_CircleNonlinear - NS NS_2D_CircleNonlinear - NFS NFS2D_CircleNonlinear

10. Circle Radius - 半径圓



Field	Value
English	Circle Radius
中文	半径圓
Formula (Math)	$x = \sin t + \cos t, y = \sin t - \cos t$
Formula (HLSL)	<code>float2(sin(InputT) + cos(InputT), sin(InputT) - cos(InputT))</code>
Description	Circle with a varying radius.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CircleRadius - NE NE_2DShapeTemplate_CircleRadius - NS NS_2D_CircleRadius - NFS NFS2D_CircleRadius

11. Cubic Curve - 三次曲线



Field	Value
English	Cubic Curve
中文	三次曲线
Formula (Math)	$x = t^3 - 3t, y = 3t^2 - 3$
Formula (HLSL)	<code>float2(InputT*InputT*InputT-3.0*InputT, 3.0*InputT*InputT-3.0)</code>
Description	Cubic curve, standard form.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CubicCurve - NE NE_2DShapeTemplate_CubicCurve - NS NS_2D_CubicCurve - NFS NFS2D_CubicCurve

12. Cubic Curve 2 - 三次曲线 2



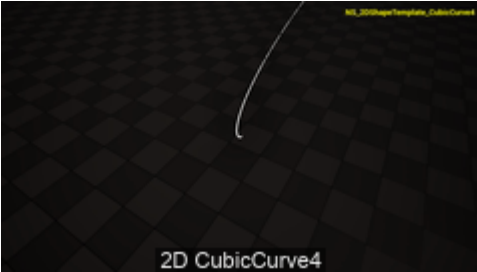
Field	Value
English	Cubic Curve 2
中文	三次曲线 2
Formula (Math)	$x = t^2 + t, y = t^3 + t$
Formula (HLSL)	<code>float2(InputT*InputT+InputT, InputT*InputT*InputT+InputT)</code>
Description	Cubic curve, variant 2.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CubicCurve2 - NE NE_2DShapeTemplate_CubicCurve2 - NS NS_2D_CubicCurve2 - NFS NFS2D_CubicCurve2

13. Cubic Curve 3 - 三次曲线 3



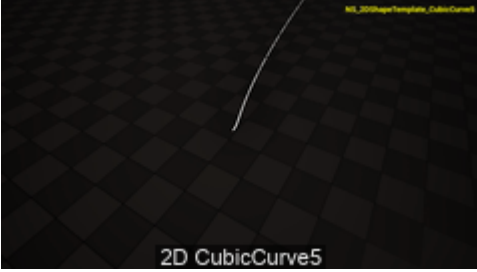
Field	Value
English	Cubic Curve 3
中文	三次曲线 3
Formula (Math)	$x = t^2 - t, y = t^3 - t^2$
Formula (HLSL)	<code>float2(InputT*InputT-InputT, InputT*InputT*InputT-InputT*InputT)</code>
Description	Cubic curve, variant 3.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CubicCurve3 - NE NE_2DShapeTemplate_CubicCurve3 - NS NS_2D_CubicCurve3 - NFS NFS2D_CubicCurve3

14. Cubic Curve 4 - 三次曲线 4



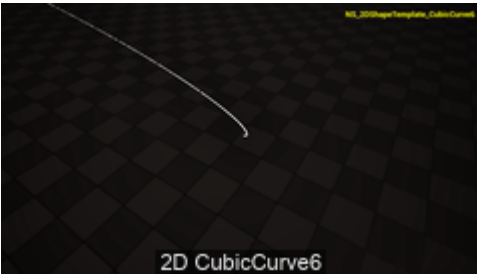
Field	Value
English	Cubic Curve 4
中文	三次曲线 4
Formula (Math)	$x = t^2, y = t^3 - t$
Formula (HLSL)	<code>float2(InputT * InputT, InputT * InputT * InputT - InputT)</code>
Description	Cubic curve, variant 4.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CubicCurve4 - NE NE_2DShapeTemplate_CubicCurve4 - NS NS_2D_CubicCurve4 - NFS NFS2D_CubicCurve4

15. Cubic Curve 5 - 三次曲线 5



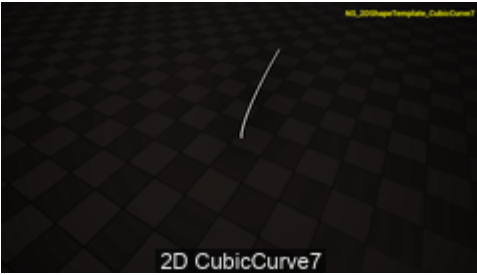
Field	Value
English	Cubic Curve 5
中文	三次曲线 5
Formula (Math)	$x = t^2 + 2, y = t^3 + 3t$
Formula (HLSL)	<code>float2(InputT * InputT + 2.0, InputT * InputT * InputT + 3.0 * InputT)</code>
Description	Cubic curve, variant 5.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CubicCurve5 - NE NE_2DShapeTemplate_CubicCurve5 - NS NS_2D_CubicCurve5 - NFS NFS2D_CubicCurve5

16. Cubic Curve 6 - 三次曲线 6



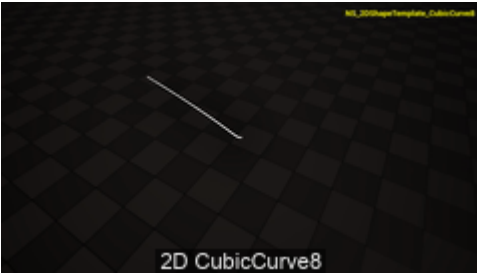
Field	Value
English	Cubic Curve 6
中文	三次曲线 6
Formula (Math)	$x = t^3 - t, y = t^2 + 1$
Formula (HLSL)	<code>float2(InputT * InputT * InputT - InputT, InputT * InputT + 1.0)</code>
Description	Cubic curve, variant 6.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CubicCurve6 - NE NE_2DShapeTemplate_CubicCurve6 - NS NS_2D_CubicCurve6 - NFS NFS2D_CubicCurve6

17. Cubic Curve 7 - 三次曲线 7



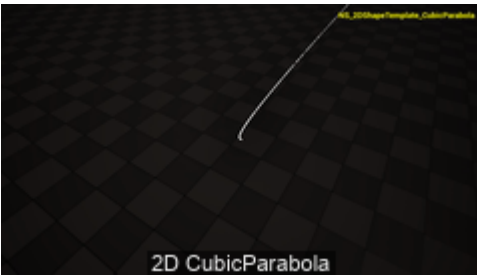
Field	Value
English	Cubic Curve 7
中文	三次曲线 7
Formula (Math)	$x = t^2, y = t^3 + t^2$
Formula (HLSL)	<code>float2(InputT * InputT, InputT * InputT * InputT + InputT * InputT)</code>
Description	Cubic curve, variant 7.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CubicCurve7 - NE NE_2DShapeTemplate_CubicCurve7 - NS NS_2D_CubicCurve7 - NFS NFS2D_CubicCurve7

18. Cubic Curve 8 - 三次曲线 8



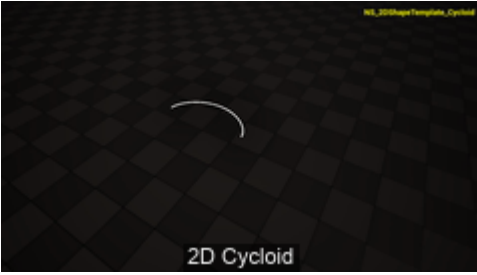
Field	Value
English	Cubic Curve 8
中文	三次曲线 8
Formula (Math)	$x = t^3 + t, y = t^2 - t$
Formula (HLSL)	<code>float2(InputT * InputT * InputT + InputT, InputT * InputT - InputT)</code>
Description	Cubic curve, variant 8.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CubicCurve8 - NE NE_2DShapeTemplate_CubicCurve8 - NS NS_2D_CubicCurve8 - NFS NFS2D_CubicCurve8

19. Cubic Parabola - 三次抛物线



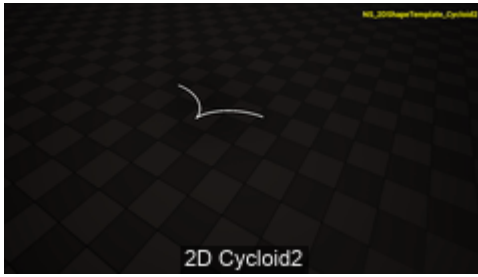
Field	Value
English	Cubic Parabola
中文	三次抛物线
Formula (Math)	$x = t, y = t^3$
Formula (HLSL)	<code>float2(InputT, InputT * InputT * InputT)</code>
Description	Cubic parabola (t, t^3).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CubicParabola - NE NE_2DShapeTemplate_CubicParabola - NS NS_2D_CubicParabola - NFS NFS2D_CubicParabola

20. Cycloid - 摆线



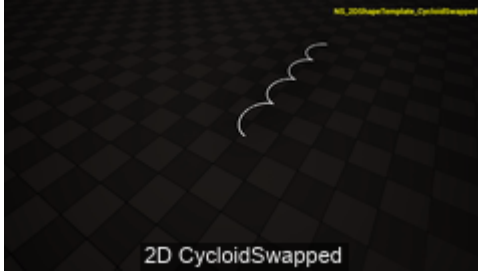
Field	Value
English	Cycloid
中文	摆线
Formula (Math)	$x = t - \sin t, y = 1 - \cos t$
Formula (HLSL)	<code>float2(InputT-sin(InputT), 1.0-cos(InputT))</code>
Description	Cycloid — path of a point on a rolling circle.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Cycloid - NE NE_2DShapeTemplate_Cycloid - NS NS_2D_Cycloid - NFS NFS2D_Cycloid

21. Cycloid 2 - 摆线 2



Field	Value
English	Cycloid 2
中文	摆线 2
Formula (Math)	$x = t + \sin t, y = 1 + \cos t$
Formula (HLSL)	<code>float2(InputT+sin(InputT), 1.0+cos(InputT))</code>
Description	Cycloid, variant with added sin/cos terms.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Cycloid2 - NE NE_2DShapeTemplate_Cycloid2 - NS NS_2D_Cycloid2 - NFS NFS2D_Cycloid2

22. Cycloid Swapped - 交换摆线



Field	Value
English	Cycloid Swapped
中文	交换摆线
Formula (Math)	$x = 1 - \cos t, y = t - \sin t$
Formula (HLSL)	<code>float2(1.0-cos(InputT), InputT-sin(InputT))</code>
Description	Cycloid with the axes swapped.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_CycloidSwapped - NE NE_2DShapeTemplate_CycloidSwapped - NS NS_2D_CycloidSwapped - NFS NFS2D_CycloidSwapped

23. Deltoid - 三角星形线



Field	Value
English	Deltoid
中文	三角星形线
Formula (Math)	$x = 2\cos t - \cos 2t, y = 2\sin t - \sin 2t$
Formula (HLSL)	<code>float2(2.0*cos(InputT)-cos(2.0*InputT), 2.0*sin(InputT)-sin(2.0*InputT))</code>
Description	Three-cusped hypocycloid (deltoid).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Deltoid - NE NE_2DShapeTemplate_Deltoid - NS NS_2D_Deltoid - NFS NFS2D_Deltoid

24. Ellipse - 椭圆



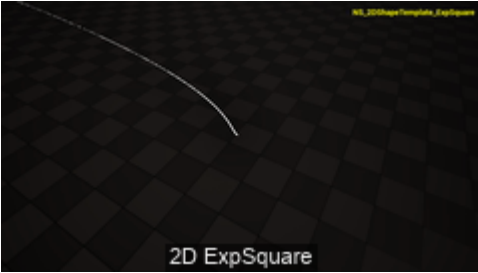
Field	Value
English	Ellipse
中文	椭圆
Formula (Math)	$x = \cos t, y = \sin t$
Formula (HLSL)	<code>float2(cos(InputT), sin(InputT))</code>
Description	Ellipse (here an axis-aligned unit circle parameterisation).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Ellipse - NE NE_2DShapeTemplate_Ellipse - NS NS_2D_Ellipse - NFS NFS2D_Ellipse

25. Equidistant - 等距曲线



Field	Value
English	Equidistant
中文	等距曲线
Formula (Math)	$x = t + \cos t, y = t + \sin t$
Formula (HLSL)	<code>float2(InputT+cos(InputT), InputT+sin(InputT))</code>
Description	Equidistant curve ($t + \cos t, t + \sin t$).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Equidistant - NE NE_2DShapeTemplate_Equidistant - NS NS_2D_Equidistant - NFS NFS2D_Equidistant

26. Exp Square - 指数平方曲线



Field	Value
English	Exp Square
中文	指数平方曲线
Formula (Math)	$x = e^t, y = t^2$
Formula (HLSL)	<code>float2(exp(InputT), InputT*InputT)</code>
Description	Exponential vs. square curve.
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_ExpSquare - NE NE_2DShapeTemplate_ExpSquare - NS NS_2D_ExpSquare - NFS NFS2D_ExpSquare

27. Folium Descartes - 笛卡尔叶形线



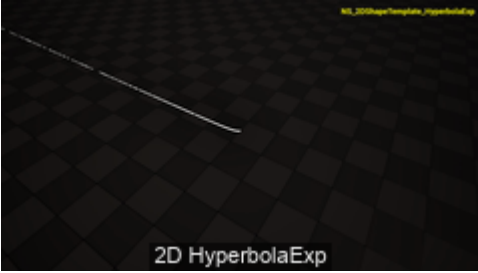
Field	Value
English	Folium Descartes
中文	笛卡尔叶形线
Formula (Math)	$x = 3t/(1 + t^3), y = 3t^2/(1 + t^3)$
Formula (HLSL)	<code>float2(3.0*InputT/(1.0+InputT*InputT*InputT), 3.0*InputT*InputT/(1.0+InputT*InputT*InputT))</code>
Description	Folium of Descartes.
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_FoliumDescartes - NE NE_2DShapeTemplate_FoliumDescartes - NS NS_2D_FoliumDescartes - NFS NFS2D_FoliumDescartes

28. Hyperbola Cosh Sinh - 双曲余弦-正弦曲线



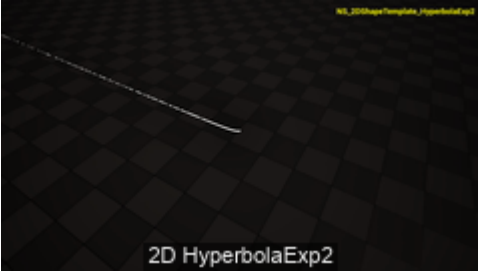
Field	Value
English	Hyperbola Cosh Sinh
中文	双曲余弦-正弦曲线
Formula (Math)	$x = \cosh t, y = \sinh t$
Formula (HLSL)	<code>float2(cosh(InputT), sinh(InputT))</code>
Description	Hyperbolic curve using cosh/sinh.
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_HyperbolaCoshSinh - NE NE_2DShapeTemplate_HyperbolaCoshSinh - NS NS_2D_HyperbolaCoshSinh - NFS NFS2D_HyperbolaCoshSinh

29. Hyperbola Exp - 指数双曲线



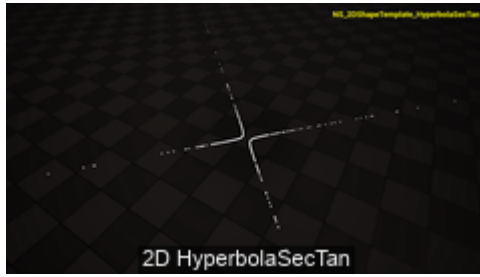
Field	Value
English	Hyperbola Exp
中文	指数双曲线
Formula (Math)	$x = e^t, y = e^{-t}$
Formula (HLSL)	<code>float2(exp(InputT), exp(-InputT))</code>
Description	Hyperbola built from exp(t) and exp(-t).
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_HyperbolaExp - NE NE_2DShapeTemplate_HyperbolaExp - NS NS_2D_HyperbolaExp - NFS NFS2D_HyperbolaExp

30. Hyperbola Exp 2 - 指数双曲线 2



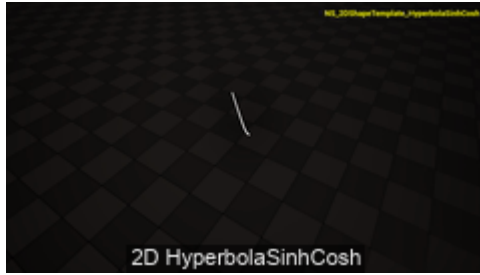
Field	Value
English	Hyperbola Exp 2
中文	指数双曲线 2
Formula (Math)	$x = e^{(t^2)}, y = e^{(-t^2)}$
Formula (HLSL)	<code>float2(exp(InputT * InputT), exp(-InputT * InputT))</code>
Description	Hyperbola built from exp(t^2) and exp(-t^2).
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_HyperbolaExp2 - NE NE_2DShapeTemplate_HyperbolaExp2 - NS NS_2D_HyperbolaExp2 - NFS NFS2D_HyperbolaExp2

31. Hyperbola Sec Tan - 正割-正切双曲线



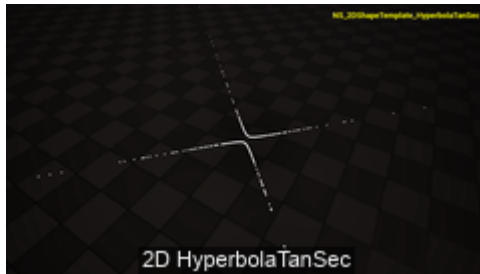
Field	Value
English	Hyperbola Sec Tan
中文	正割-正切双曲线
Formula (Math)	$x = \sec t, y = \tan t$
Formula (HLSL)	<code>float2(1.0/cos(InputT), tan(InputT))</code>
Description	Hyperbolic curve using sec/tan.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_HyperbolaSecTan - NE NE_2DShapeTemplate_HyperbolaSecTan - NS NS_2D_HyperbolaSecTan - NFS NFS2D_HyperbolaSecTan

32. Hyperbola Sinh Cosh - 双曲正弦-余弦曲线



Field	Value
English	Hyperbola Sinh Cosh
中文	双曲正弦-余弦曲线
Formula (Math)	$x = \sinh t, y = \cosh t$
Formula (HLSL)	<code>float2(sinh(InputT), cosh(InputT))</code>
Description	Hyperbolic curve using sinh/cosh.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_HyperbolaSinhCosh - NE NE_2DShapeTemplate_HyperbolaSinhCosh - NS NS_2D_HyperbolaSinhCosh - NFS NFS2D_HyperbolaSinhCosh

33. Hyperbola Tan Sec - 正切-正割双曲线



Field	Value
English	Hyperbola Tan Sec
中文	正切-正割双曲线
Formula (Math)	$x = \tan t, y = \sec t$
Formula (HLSL)	<code>float2(tan(InputT), 1.0 / cos(InputT))</code>
Description	Hyperbolic curve using tan/sec.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_HyperbolaTanSec - NE NE_2DShapeTemplate_HyperbolaTanSec - NS NS_2D_HyperbolaTanSec - NFS NFS2D_HyperbolaTanSec

34. Involute Circle - 圆的渐开线



Field	Value
English	Involute Circle
中文	圆的渐开线
Formula (Math)	$x = \cos t + t \sin t, y = \sin t - t \cos t$
Formula (HLSL)	<code>float2(cos(InputT)+InputT*sin(InputT), sin(InputT)-InputT*cos(InputT))</code>
Description	Involute of a circle.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_InvoluteCircle - NE NE_2DShapeTemplate_InvoluteCircle - NS NS_2D_InvoluteCircle - NFS NFS2D_InvoluteCircle

35. Line Segment - 线段



Field	Value
English	Line Segment
中文	线段
Formula (Math)	$x = \arcsin t, y = \arccos t$
Formula (HLSL)	<code>float2(asin(InputT), acos(InputT))</code>
Description	Line segment (asin/acos parameterisation).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_LineSegment - NE NE_2DShapeTemplate_LineSegment - NS NS_2D_LineSegment - NFS NFS2D_LineSegment

36. Lissajous 12 - 利萨茹曲线 1:2



Field	Value
English	Lissajous 12
中文	利萨茹曲线 1:2
Formula (Math)	$x = \sin t, y = \sin 2t$
Formula (HLSL)	<code>float2(sin(InputT), sin(2.0*InputT))</code>
Description	Lissajous figure, frequency ratio 1:2.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Lissajous12 - NE NE_2DShapeTemplate_Lissajous12 - NS NS_2D_Lissajous12 - NFS NFS2D_Lissajous12

37. Lissajous 12 Cos - 利萨茹曲线 1:2 (余弦)



Field	Value
English	Lissajous 12 Cos
中文	利萨茹曲线 1:2 (余弦)
Formula (Math)	$x = \sin t, y = \cos 2t$
Formula (HLSL)	<code>float2(sin(InputT), cos(2.0*InputT))</code>
Description	Lissajous figure 1:2 using cosine.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Lissajous12Cos - NE NE_2DShapeTemplate_Lissajous12Cos - NS NS_2D_Lissajous12Cos - NFS NFS2D_Lissajous12Cos

38. Lissajous 13 - 利萨茹曲线 1:3



Field	Value
English	Lissajous 13
中文	利萨茹曲线 1:3
Formula (Math)	$x = \sin t, y = \cos 3t$
Formula (HLSL)	<code>float2(sin(InputT), cos(3.0*InputT))</code>
Description	Lissajous figure, frequency ratio 1:3.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Lissajous13 - NE NE_2DShapeTemplate_Lissajous13 - NS NS_2D_Lissajous13 - NFS NFS2D_Lissajous13

39. Lissajous 21 - 利萨茹曲线 2:1



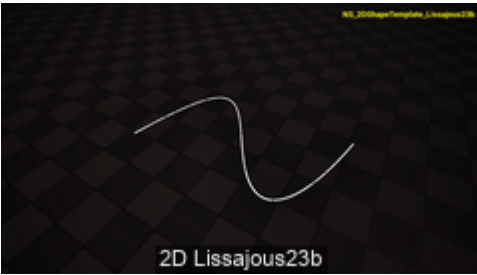
Field	Value
English	Lissajous 21
中文	利萨茹曲线 2:1
Formula (Math)	$x = \sin 2t, y = \sin t$
Formula (HLSL)	<code>float2(sin(2.0*InputT), sin(InputT))</code>
Description	Lissajous figure, frequency ratio 2:1.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Lissajous21 - NE NE_2DShapeTemplate_Lissajous21 - NS NS_2D_Lissajous21 - NFS NFS2D_Lissajous21

40. Lissajous 23 - 利萨茹曲线 2:3



Field	Value
English	Lissajous 23
中文	利萨茹曲线 2:3
Formula (Math)	$x = \cos 2t, y = \cos 3t$
Formula (HLSL)	<code>float2(cos(2.0*InputT), cos(3.0*InputT))</code>
Description	Lissajous figure, frequency ratio 2:3.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Lissajous23 - NE NE_2DShapeTemplate_Lissajous23 - NS NS_2D_Lissajous23 - NFS NFS2D_Lissajous23

41. Lissajous 23b - 利萨茹曲线 2:3 变体 B



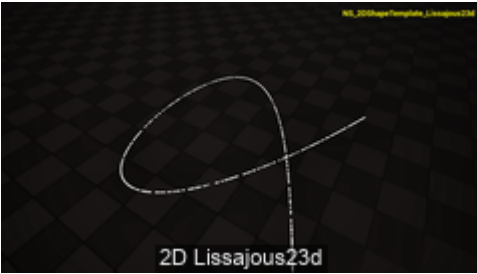
Field	Value
English	Lissajous 23b
中文	利萨茹曲线 2:3 变体 B
Formula (Math)	$x = 2\sin t, y = \sin 3t$
Formula (HLSL)	<code>float2(2.0*sin(InputT), sin(3.0*InputT))</code>
Description	Lissajous figure 2:3, variant B.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Lissajous23b - NE NE_2DShapeTemplate_Lissajous23b - NS NS_2D_Lissajous23b - NFS NFS2D_Lissajous23b

42. Lissajous 23c - 利萨茹曲线 2:3 变体 C



Field	Value
English	Lissajous 23c
中文	利萨茹曲线 2:3 变体 C
Formula (Math)	$x = \sin 2t, y = \sin 3t$
Formula (HLSL)	<code>float2(sin(2.0*InputT), sin(3.0*InputT))</code>
Description	Lissajous figure 2:3, variant C.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Lissajous23c - NE NE_2DShapeTemplate_Lissajous23c - NS NS_2D_Lissajous23c - NFS NFS2D_Lissajous23c

43. Lissajous 23d - 利萨茹曲线 2:3 变体 D



Field	Value
English	Lissajous 23d
中文	利萨茹曲线 2:3 变体 D
Formula (Math)	$x = \cos 2t, y = \sin 3t$
Formula (HLSL)	<code>float2(cos(2.0 * InputT), sin(3.0 * InputT))</code>
Description	Lissajous figure 2:3, variant D.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Lissajous23d - NE NE_2DShapeTemplate_Lissajous23d - NS NS_2D_Lissajous23d - NFS NFS2D_Lissajous23d

44. Lissajous 32 - 利萨茹曲线 3:2



Field	Value
English	Lissajous 32
中文	利萨茹曲线 3:2
Formula (Math)	$x = \sin 3t, y = \sin 2t$
Formula (HLSL)	<code>float2(sin(3.0*InputT), sin(2.0*InputT))</code>
Description	Lissajous figure, frequency ratio 3:2.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Lissajous32 - NE NE_2DShapeTemplate_Lissajous32 - NS NS_2D_Lissajous32 - NFS NFS2D_Lissajous32

45. Lissajous 34 - 利萨茹曲线 3:4



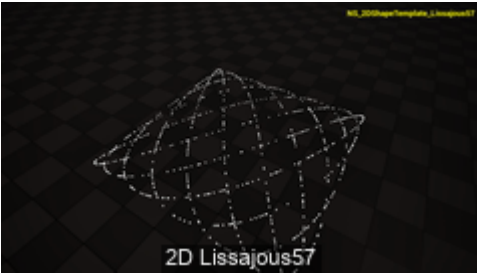
Field	Value
English	Lissajous 34
中文	利萨茹曲线 3:4
Formula (Math)	$x = \cos 3t, y = \sin 4t$
Formula (HLSL)	<code>float2(cos(3.0 * InputT), sin(4.0 * InputT))</code>
Description	Lissajous figure, frequency ratio 3:4.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Lissajous34 - NE NE_2DShapeTemplate_Lissajous34 - NS NS_2D_Lissajous34 - NFS NFS2D_Lissajous34

46. Lissajous 45 - 利萨茹曲线 4:5



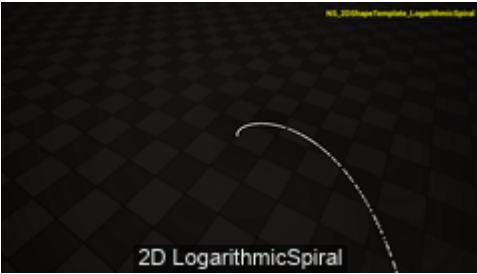
Field	Value
English	Lissajous 45
中文	利萨茹曲线 4:5
Formula (Math)	$x = \cos 4t, y = \sin 5t$
Formula (HLSL)	<code>float2(cos(4.0*InputT), sin(5.0*InputT))</code>
Description	Lissajous figure, frequency ratio 4:5.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_Lissajous45 - NE NE_2DShapeTemplate_Lissajous45 - NS NS_2D_Lissajous45 - NFS NFS2D_Lissajous45

47. Lissajous 57 - 利萨茹曲线 5:7



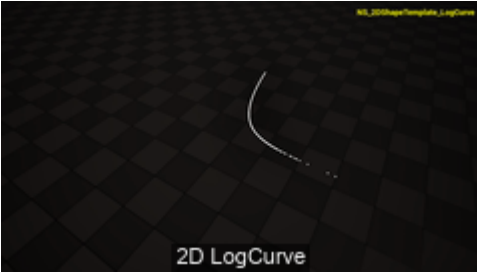
Field	Value
English	Lissajous 57
中文	利萨茹曲线 5:7
Formula (Math)	$x = \sin 5t, y = \cos 7t$
Formula (HLSL)	<code>float2(sin(5.0 * InputT), cos(7.0 * InputT))</code>
Description	Lissajous figure, frequency ratio 5:7.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_Lissajous57 - NE NE_2DShapeTemplate_Lissajous57 - NS NS_2D_Lissajous57 - NFS NFS2D_Lissajous57

48. Logarithmic Spiral - 对数螺线



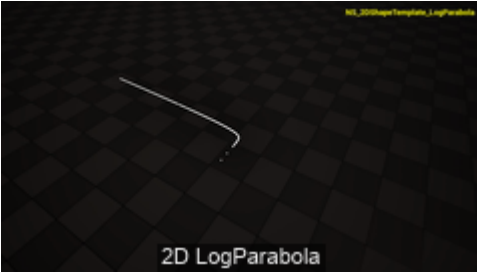
Field	Value
English	Logarithmic Spiral
中文	对数螺线
Formula (Math)	$x = e^t \cos t, y = e^t \sin t$
Formula (HLSL)	<code>float2(exp(InputT) * cos(InputT), exp(InputT) * sin(InputT))</code>
Description	Logarithmic (equiangular) spiral.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_LogarithmicSpiral - NE NE_2DShapeTemplate_LogarithmicSpiral - NS NS_2D_LogarithmicSpiral - NFS NFS2D_LogarithmicSpiral

49. Log Curve - 对数曲线



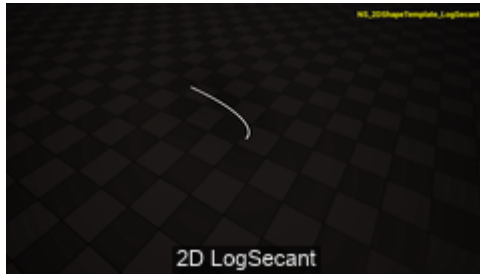
Field	Value
English	Log Curve
中文	对数曲线
Formula (Math)	$x = \log t, y = t$
Formula (HLSL)	<code>float2(Log(InputT), InputT)</code>
Description	Logarithmic curve (log t, t).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_LogCurve - NE NE_2DShapeTemplate_LogCurve - NS NS_2D_LogCurve - NFS NFS2D_LogCurve

50. Log Parabola - 对数抛物线



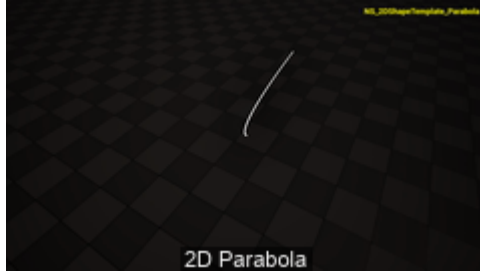
Field	Value
English	Log Parabola
中文	对数抛物线
Formula (Math)	$x = t^2, y = \log t$
Formula (HLSL)	<code>float2(InputT*InputT, log(InputT))</code>
Description	Log vs. parabolic curve (t^2 , log t).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_LogParabola - NE NE_2DShapeTemplate_LogParabola - NS NS_2D_LogParabola - NFS NFS2D_LogParabola

51. Log Secant - 对数正割曲线



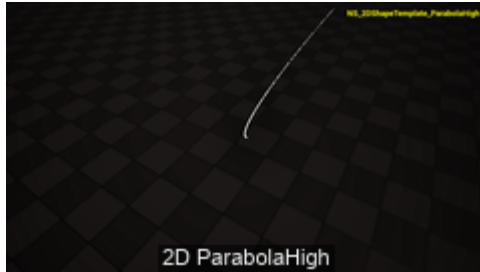
Field	Value
English	Log Secant
中文	对数正割曲线
Formula (Math)	$x = \log(1 + t^2), y = \arctan t$
Formula (HLSL)	<code>float2(Log(InputT * InputT + 1.0), atan(InputT))</code>
Description	Log/arctangent curve.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_LogSecant - NE NE_2DShapeTemplate_LogSecant - NS NS_2D_LogSecant - NFS NFS2D_LogSecant

52. Parabola - 抛物线



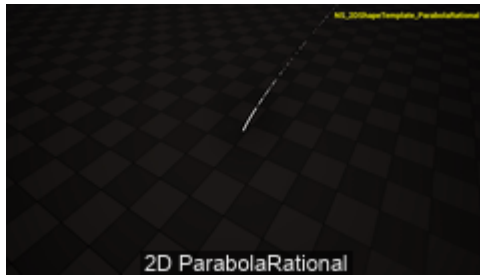
Field	Value
English	Parabola
中文	抛物线
Formula (Math)	$x = t, y = t^2$
Formula (HLSL)	<code>float2(InputT, InputT * InputT)</code>
Description	Parabola (t, t^2).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_Parabola - NE NE_2DShapeTemplate_Parabola - NS NS_2D_Parabola - NFS NFS2D_Parabola

53. Parabola High - 高次抛物线



Field	Value
English	Parabola High
中文	高次抛物线
Formula (Math)	$x = t^5, y = t^{10}$
Formula (HLSL)	<code>float2(pow(InputT, 5.0), pow(InputT, 10.0))</code>
Description	High-order parabola (t^5, t^{10}).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_ParabolaHigh - NE NE_2DShapeTemplate_ParabolaHigh - NS NS_2D_ParabolaHigh - NFS NFS2D_ParabolaHigh

54. Parabola Rational - 有理抛物线



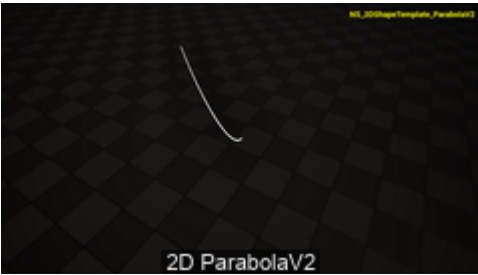
Field	Value
English	Parabola Rational
中文	有理抛物线
Formula (Math)	$x = t + 1/t, y = t^2 + 1/t^2$
Formula (HLSL)	<code>float2(InputT + 1.0 / InputT, InputT * InputT + 1.0 / (InputT * InputT))</code>
Description	Rational parabola with $1/t$ terms.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_ParabolaRational - NE NE_2DShapeTemplate_ParabolaRational - NS NS_2D_ParabolaRational - NFS NFS2D_ParabolaRational

55. Parabola Shifted - 平移抛物线



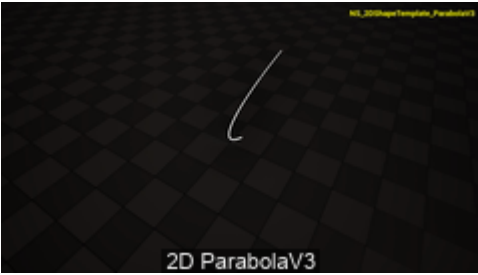
Field	Value
English	Parabola Shifted
中文	平移抛物线
Formula (Math)	$x = t + 1, y = (t + 1)^2$
Formula (HLSL)	<code>float2(InputT + 1.0, (InputT + 1.0) * (InputT + 1.0))</code>
Description	Parabola shifted by +1 on both axes.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_ParabolaShifted - NE NE_2DShapeTemplate_ParabolaShifted - NS NS_2D_ParabolaShifted - NFS NFS2D_ParabolaShifted

56. Parabola V2 - 抛物线 V2



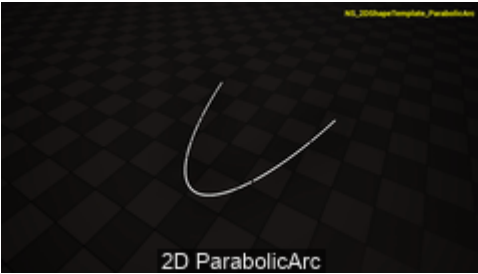
Field	Value
English	Parabola V2
中文	抛物线 V2
Formula (Math)	$x = t^2, y = t^2 - 2t$
Formula (HLSL)	<code>float2(InputT*InputT, InputT*InputT-2.0*InputT)</code>
Description	Parabola, variant 2.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_ParabolaV2 - NE NE_2DShapeTemplate_ParabolaV2 - NS NS_2D_ParabolaV2 - NFS NFS2D_ParabolaV2

57. Parabola V3 - 抛物线 V3



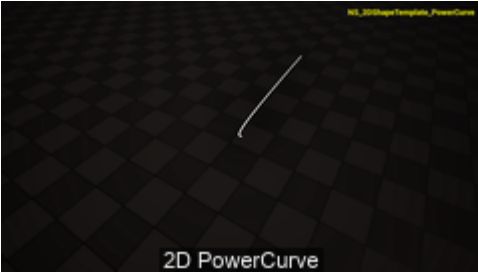
Field	Value
English	Parabola V3
中文	抛物线 V3
Formula (Math)	$x = t, y = t^2 - 2t$
Formula (HLSL)	<code>float2(InputT, InputT*InputT-2.0*InputT)</code>
Description	Parabola, variant 3.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_ParabolaV3 - NE NE_2DShapeTemplate_ParabolaV3 - NS NS_2D_ParabolaV3 - NFS NFS2D_ParabolaV3

58. Parabolic Arc - 抛物弧线



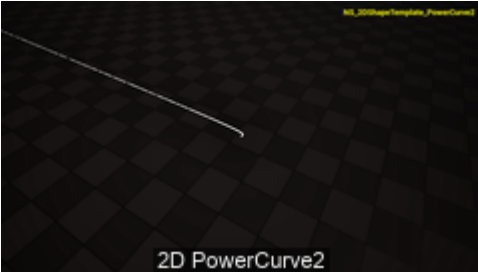
Field	Value
English	Parabolic Arc
中文	抛物弧线
Formula (Math)	$x = \cos t, y = \cos 2t$
Formula (HLSL)	<code>float2(cos(InputT), cos(2.0*InputT))</code>
Description	Parabolic arc (cos t, cos 2t).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_ParabolicArc - NE NE_2DShapeTemplate_ParabolicArc - NS NS_2D_ParabolicArc - NFS NFS2D_ParabolicArc

59. Power Curve - 幂曲线



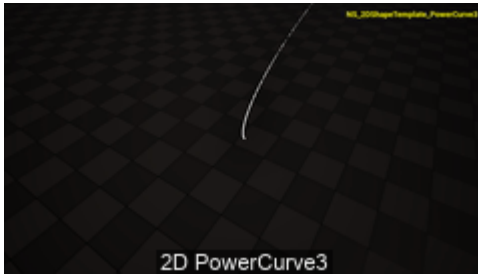
Field	Value
English	Power Curve
中文	幂曲线
Formula (Math)	$x = \sqrt{t}, y = t^2$
Formula (HLSL)	<code>float2(sqrt(InputT), InputT*InputT)</code>
Description	Power curve (sqrt t, t^2).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_PowerCurve - NE NE_2DShapeTemplate_PowerCurve - NS NS_2D_PowerCurve - NFS NFS2D_PowerCurve

60. Power Curve 2 - 幂曲线 2



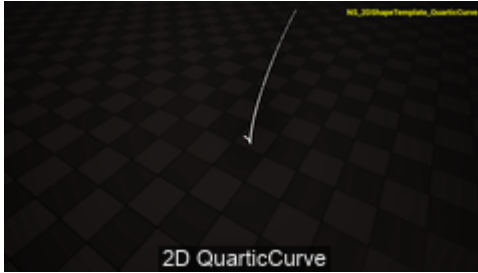
Field	Value
English	Power Curve 2
中文	幂曲线 2
Formula (Math)	$x = t^3, y = \sqrt{t}$
Formula (HLSL)	<code>float2(InputT*InputT*InputT, sqrt(InputT))</code>
Description	Power curve (t^3, sqrt t).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_PowerCurve2 - NE NE_2DShapeTemplate_PowerCurve2 - NS NS_2D_PowerCurve2 - NFS NFS2D_PowerCurve2

61. Power Curve 3 - 幂曲线 3



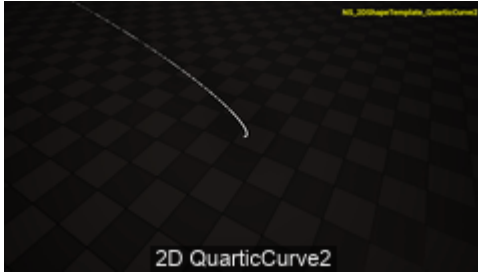
Field	Value
English	Power Curve 3
中文	幂曲线 3
Formula (Math)	$x = t^4, y = t^6$
Formula (HLSL)	<code>float2(pow(InputT, 4.0), pow(InputT, 6.0))</code>
Description	Power curve (t^4, t^6).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_PowerCurve3 - NE NE_2DShapeTemplate_PowerCurve3 - NS NS_2D_PowerCurve3 - NFS NFS2D_PowerCurve3

62. Quartic Curve - 四次曲线



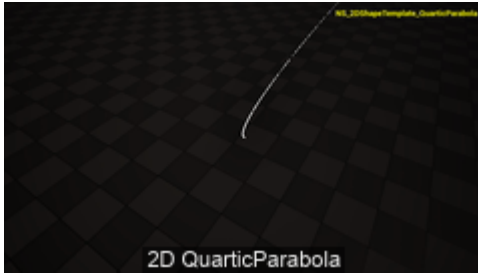
Field	Value
English	Quartic Curve
中文	四次曲线
Formula (Math)	$x = t^3 - 3t, y = t^4 - 2t^2$
Formula (HLSL)	<code>float2(InputT * InputT * InputT - 3.0 * InputT, pow(InputT, 4.0) - 2.0 * InputT * InputT)</code>
Description	Quartic curve.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_QuarticCurve - NE NE_2DShapeTemplate_QuarticCurve - NS NS_2D_QuarticCurve - NFS NFS2D_QuarticCurve

63. Quartic Curve 2 - 四次曲线 2



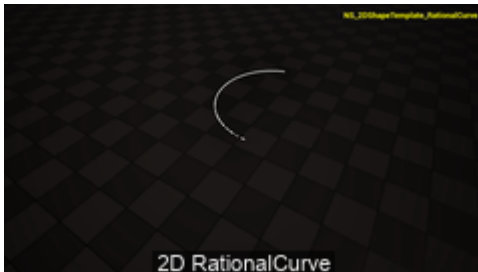
Field	Value
English	Quartic Curve 2
中文	四次曲线 2
Formula (Math)	$x = t^4 - t^2, y = t^3$
Formula (HLSL)	<code>float2(pow(InputT, 4.0) - InputT * InputT, InputT * InputT * InputT)</code>
Description	Quartic curve, variant 2.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_QuarticCurve2 - NE NE_2DShapeTemplate_QuarticCurve2 - NS NS_2D_QuarticCurve2 - NFS NFS2D_QuarticCurve2

64. Quartic Parabola - 四次抛物线



Field	Value
English	Quartic Parabola
中文	四次抛物线
Formula (Math)	$x = t^2, y = t^4$
Formula (HLSL)	<code>float2(InputT * InputT, pow(InputT, 4.0))</code>
Description	Quartic parabola (t^2, t^4).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_QuarticParabola - NE NE_2DShapeTemplate_QuarticParabola - NS NS_2D_QuarticParabola - NFS NFS2D_QuarticParabola

65. Rational Curve - 有理曲线



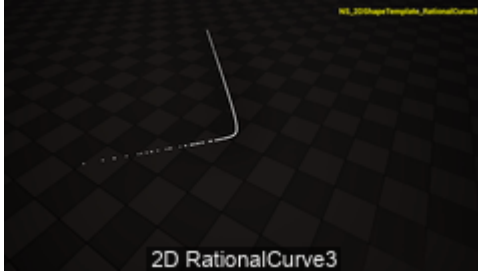
Field	Value
English	Rational Curve
中文	有理曲线
Formula (Math)	$x = t/(1 + t^2), y = t^2/(1 + t^2)$
Formula (HLSL)	<code>float2(InputT/(1.0+InputT*InputT), InputT*InputT/(1.0+InputT*InputT))</code>
Description	Rational curve.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_RationalCurve - NE NE_2DShapeTemplate_RationalCurve - NS NS_2D_RationalCurve - NFS NFS2D_RationalCurve

66. Rational Curve 2 - 有理曲线 2



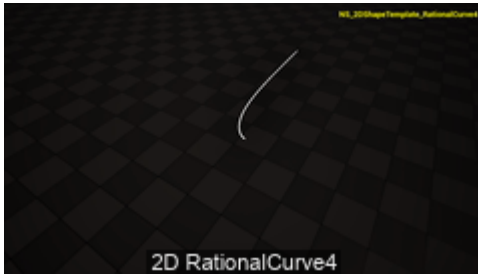
Field	Value
English	Rational Curve 2
中文	有理曲线 2
Formula (Math)	$x = t^2/(1 + t^2), y = t/(1 + t^2)$
Formula (HLSL)	<code>float2(InputT*InputT/(1.0+InputT*InputT), InputT/(1.0+InputT*InputT))</code>
Description	Rational curve, variant 2.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_RationalCurve2 - NE NE_2DShapeTemplate_RationalCurve2 - NS NS_2D_RationalCurve2 - NFS NFS2D_RationalCurve2

67. Rational Curve 3 - 有理曲线 3



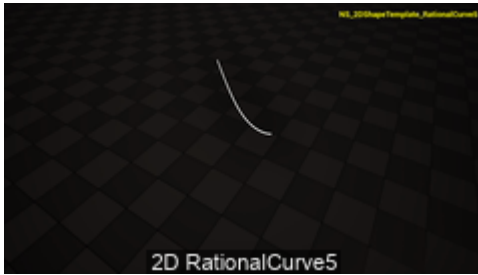
Field	Value
English	Rational Curve 3
中文	有理曲线 3
Formula (Math)	$x = t^2 + 1/t, y = t^2 - 1/t$
Formula (HLSL)	<code>float2(InputT*InputT+1.0/InputT, InputT*InputT-1.0/InputT)</code>
Description	Rational curve, variant 3.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_RationalCurve3 - NE NE_2DShapeTemplate_RationalCurve3 - NS NS_2D_RationalCurve3 - NFS NFS2D_RationalCurve3

68. Rational Curve 4 - 有理曲线 4



Field	Value
English	Rational Curve 4
中文	有理曲线 4
Formula (Math)	$x = t^2/(1 + t^2), y = t^3/(1 + t^2)$
Formula (HLSL)	<code>float2(InputT*InputT/(1.0+InputT*InputT), InputT*InputT*InputT/(1.0+InputT*InputT))</code>
Description	Rational curve, variant 4.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_RationalCurve4 - NE NE_2DShapeTemplate_RationalCurve4 - NS NS_2D_RationalCurve4 - NFS NFS2D_RationalCurve4

69. Rational Curve 5 - 有理曲线 5



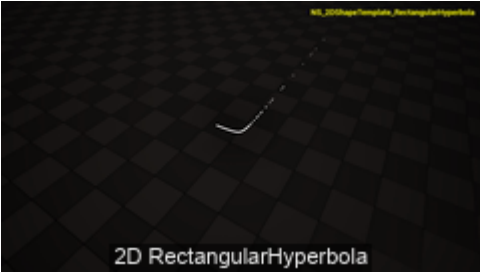
Field	Value
English	Rational Curve 5
中文	有理曲线 5
Formula (Math)	$x = (t^3 - 1)/(t^2 + 1), y = (t^3 + 1)/(t^2 + 1)$
Formula (HLSL)	<code>float2((InputT*InputT*InputT-1.0)/(InputT*InputT+1.0), (InputT*InputT*InputT+1.0)/(InputT*InputT+1.0))</code>
Description	Rational curve, variant 5.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_RationalCurve5 - NE NE_2DShapeTemplate_RationalCurve5 - NS NS_2D_RationalCurve5 - NFS NFS2D_RationalCurve5

70. Rational Curve 6 - 有理曲线 6



Field	Value
English	Rational Curve 6
中文	有理曲线 6
Formula (Math)	$x = t^2 + 1/t^2, y = t^3 + 1/t^3$
Formula (HLSL)	<code>float2(InputT * InputT + 1.0 / (InputT * InputT), InputT * InputT * InputT + 1.0 / (InputT * InputT * InputT))</code>
Description	Rational curve, variant 6.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_2D_RationalCurve6 - NE NE_2DShapeTemplate_RationalCurve6 - NS NS_2D_RationalCurve6 - NFS NFS2D_RationalCurve6

71. Rectangular Hyperbola - 直角双曲线



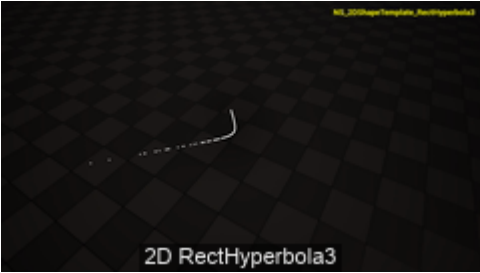
Field	Value
English	Rectangular Hyperbola
中文	直角双曲线
Formula (Math)	$x = t, y = 1/t$
Formula (HLSL)	<code>float2(InputT, 1.0/InputT)</code>
Description	Rectangular hyperbola (t, 1/t).
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_RectangularHyperbola - NE NE_2DShapeTemplate_RectangularHyperbola - NS NS_2D_RectangularHyperbola - NFS NFS2D_RectangularHyperbola

72. Rect Hyperbola 2 - 直角双曲线 2



Field	Value
English	Rect Hyperbola 2
中文	直角双曲线 2
Formula (Math)	$x = t + 1/t, y = t - 1/t$
Formula (HLSL)	<code>float2(InputT+1.0/InputT, InputT-1.0/InputT)</code>
Description	Rectangular hyperbola, variant 2.
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_RectHyperbola2 - NE NE_2DShapeTemplate_RectHyperbola2 - NS NS_2D_RectHyperbola2 - NFS NFS2D_RectHyperbola2

73. Rect Hyperbola 3 - 直角双曲线 3



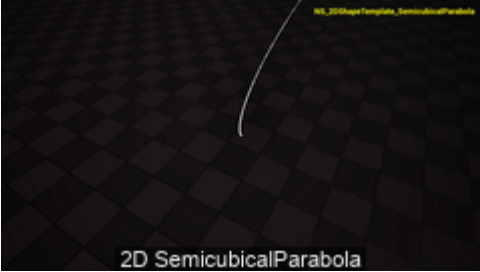
Field	Value
English	Rect Hyperbola 3
中文	直角双曲线 3
Formula (Math)	$x = (t^2 + 1)/t, y = (t^2 - 1)/t$
Formula (HLSL)	<code>float2((InputT * InputT + 1.0) / InputT, (InputT * InputT - 1.0) / InputT)</code>
Description	Rectangular hyperbola, variant 3.
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_RectHyperbola3 - NE NE_2DShapeTemplate_RectHyperbola3 - NS NS_2D_RectHyperbola3 - NFS NFS2D_RectHyperbola3

74. Rotated Ellipse - 旋转椭圆



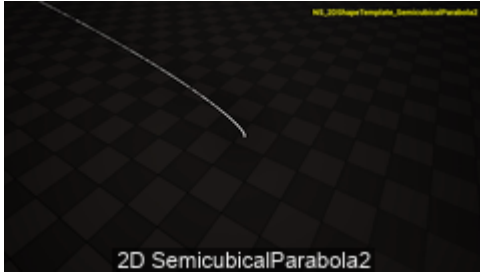
Field	Value
English	Rotated Ellipse
中文	旋转椭圆
Formula (Math)	$x = \sin t, y = \sin(t + \pi/3)$
Formula (HLSL)	<code>float2(sin(InputT), sin(InputT + 1.0472))</code>
Description	Rotated ellipse (sin t, sin(t + 60°)).
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_RotatedEllipse - NE NE_2DShapeTemplate_RotatedEllipse - NS NS_2D_RotatedEllipse - NFS NFS2D_RotatedEllipse

75. Semicubical Parabola - 半立方抛物线



Field	Value
English	Semicubical Parabola
中文	半立方抛物线
Formula (Math)	$x = t^2, y = t^3$
Formula (HLSL)	<code>float2(InputT * InputT, InputT * InputT * InputT)</code>
Description	Semicubical parabola (t^2, t^3).
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_SemicubicalParabola - NE NE_2DShapeTemplate_SemicubicalParabola - NS NS_2D_SemicubicalParabola - NFS NFS2D_SemicubicalParabola

76. Semicubical Parabola 2 - 半立方抛物线 2



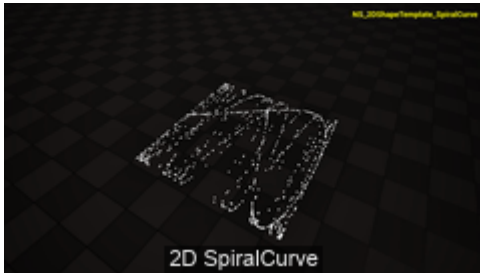
Field	Value
English	Semicubical Parabola 2
中文	半立方抛物线 2
Formula (Math)	$x = t^3, y = t^2$
Formula (HLSL)	<code>float2(InputT*InputT*InputT, InputT*InputT)</code>
Description	Semicubical parabola, axes swapped.
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_SemicubicalParabola2 - NE NE_2DShapeTemplate_SemicubicalParabola2 - NS NS_2D_SemicubicalParabola2 - NFS NFS2D_SemicubicalParabola2

77. Sine Squared - 正弦平方曲线



Field	Value
English	Sine Squared
中文	正弦平方曲线
Formula (Math)	$x = \cos t, y = \sin^2 t$
Formula (HLSL)	<code>float2(cos(InputT), sin(InputT)*sin(InputT))</code>
Description	Sine-squared curve ($\cos t, \sin^2 t$).
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_SineSquared - NE NE_2DShapeTemplate_SineSquared - NS NS_2D_SineSquared - NFS NFS2D_SineSquared

78. Spiral Curve - 螺旋曲线



Field	Value
English	Spiral Curve
中文	螺旋曲线
Formula (Math)	$x = \cos(\pi t), y = \sin(\pi^2 t)$
Formula (HLSL)	<code>float2(cos(3.14159 * InputT), sin(3.14159 * InputT * InputT))</code>
Description	Spiral curve ($\cos \pi t, \sin \pi^2 t$).
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_SpiralCurve - NE NE_2DShapeTemplate_SpiralCurve - NS NS_2D_SpiralCurve - NFS NFS2D_SpiralCurve

79. Tractrix - 曳物线



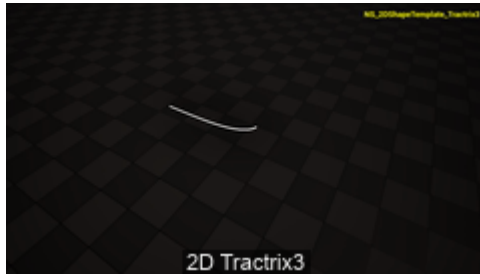
Field	Value
English	Tractrix
中文	曳物线
Formula (Math)	$x = \cos t + \log \tan(t/2), y = \sin t$
Formula (HLSL)	<code>float2(cos(InputT)+log(tan(InputT*0.5)), sin(InputT))</code>
Description	Tractrix (pursuit curve).
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_Tractrix - NE NE_2DShapeTemplate_Tractrix - NS NS_2D_Tractrix - NFS NFS2D_Tractrix

80. Tractrix 2 - 曳物线 2



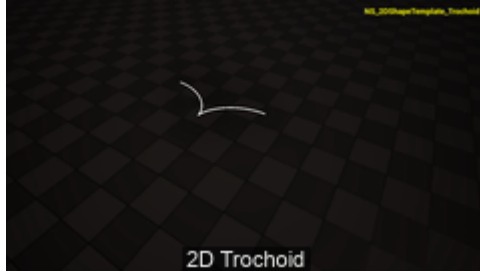
Field	Value
English	Tractrix 2
中文	曳物线 2
Formula (Math)	$x = 1/\cosh t, y = t - \tanh t$
Formula (HLSL)	<code>float2(1.0/cosh(InputT), InputT-tanh(InputT))</code>
Description	Tractrix, variant 2.
Type / Output	2D parametric curve / <code>float2</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_Tractrix2 - NE NE_2DShapeTemplate_Tractrix2 - NS NS_2D_Tractrix2 - NFS NFS2D_Tractrix2

81. Tractrix 3 - 曳物线 3



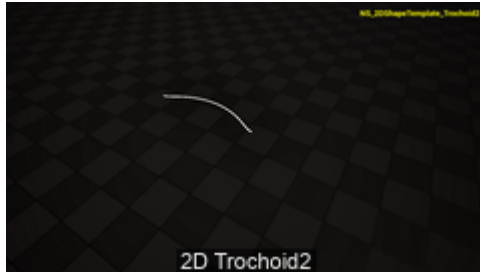
Field	Value
English	Tractrix 3
中文	曳物线 3
Formula (Math)	$x = t - \tanh t, y = 1/\cosh t$
Formula (HLSL)	<code>float2(InputT - tanh(InputT), 1.0 / cosh(InputT))</code>
Description	Tractrix, variant 3.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_Tractrix3 - NE NE_2DShapeTemplate_Tractrix3 - NS NS_2D_Tractrix3 - NFS NFS2D_Tractrix3

82. Trochoid - 次摆线



Field	Value
English	Trochoid
中文	次摆线
Formula (Math)	$x = t + \sin t, y = 1 + \cos t$
Formula (HLSL)	<code>float2(InputT+sin(InputT), 1.0+cos(InputT))</code>
Description	Trochoid (extended cycloid).
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_Trochoid - NE NE_2DShapeTemplate_Trochoid - NS NS_2D_Trochoid - NFS NFS2D_Trochoid

83. Trochoid 2 - 次摆线 2



Field	Value
English	Trochoid 2
中文	次摆线 2
Formula (Math)	$x = 2t - \sin t, y = 2 - \cos t$
Formula (HLSL)	<code>float2(2.0*InputT-sin(InputT), 2.0-cos(InputT))</code>
Description	Trochoid, variant 2.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_Trochoid2 - NE NE_2DShapeTemplate_Trochoid2 - NS NS_2D_Trochoid2 - NFS NFS2D_Trochoid2

84. Unit Circle - 单位圆



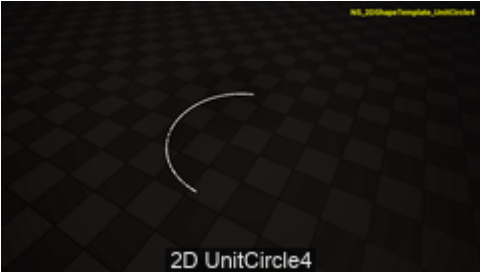
Field	Value
English	Unit Circle
中文	单位圆
Formula (Math)	$x = (1 - t^2)/(1 + t^2), y = 2t/(1 + t^2)$
Formula (HLSL)	<code>float2((1.0-InputT*InputT)/(1.0+InputT*InputT), 2.0*InputT/(1.0+InputT*InputT))</code>
Description	Rational unit circle.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_UnitCircle - NE NE_2DShapeTemplate_UnitCircle - NS NS_2D_UnitCircle - NFS NFS2D_UnitCircle

85. Unit Circle 3 - 单位圆 3



Field	Value
English	Unit Circle 3
中文	单位圆 3
Formula (Math)	$x = (t^2 - 1)/(t^2 + 1), y = 2t/(t^2 + 1)$
Formula (HLSL)	<code>float2((InputT * InputT - 1.0) / (InputT * InputT + 1.0), 2.0 * InputT / (InputT * InputT + 1.0))</code>
Description	Rational unit circle, variant 3.
Type / Output	2D parametric curve / float2
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_2D_UnitCircle3 - NE NE_2DShapeTemplate_UnitCircle3 - NS NS_2D_UnitCircle3 - NFS NFS2D_UnitCircle3

86. Unit Circle 4 - 单位圆 4



Field	Value
English	Unit Circle 4
中文	单位圆 4
Formula (Math)	$x = 2t/(t^2 + 1), y = (t^2 - 1)/(t^2 + 1)$
Formula (HLSL)	<code>float2(2.0 * InputT / (InputT * InputT + 1.0), (InputT * InputT - 1.0) / (InputT * InputT + 1.0))</code>
Description	Rational unit circle, variant 4.
Type / Output	2D parametric curve / float2
Input	Module.InputT (float)
Assets	NDI NDI_2D_UnitCircle4 - NE NE_2DShapeTemplate_UnitCircle4 - NS NS_2D_UnitCircle4 - NFS NFS2D_UnitCircle4

87. Unit Circle Hyperbolic - 双曲单位圆



Field	Value
English	Unit Circle Hyperbolic
中文	双曲单位圆
Formula (Math)	$x = \tanh t, y = 1/\cosh t$
Formula (HLSL)	<code>float2(tanh(InputT), 1.0 / cosh(InputT))</code>
Description	Hyperbolic unit circle (tanh, sech).
Type / Output	2D parametric curve / float2
Input	Module.InputT (float)
Assets	NDI NDI_2D_UnitCircleHyperbolic - NE NE_2DShapeTemplate_UnitCircleHyperbolic - NS NS_2D_UnitCircleHyperbolic - NFS NFS2D_UnitCircleHyperbolic

88. Witch Agnesi - 阿涅西箕舌线



Field	Value
English	Witch Agnesi
中文	阿涅西箕舌线
Formula (Math)	$x = t, y = 1/(1 + t^2)$
Formula (HLSL)	<code>float2(InputT, 1.0 / (1.0 + InputT * InputT))</code>
Description	Witch of Agnesi.
Type / Output	2D parametric curve / float2
Input	Module.InputT (float)
Assets	NDI NDI_2D_WitchAgnesi - NE NE_2DShapeTemplate_WitchAgnesi - NS NS_2D_WitchAgnesi - NFS NFS2D_WitchAgnesi

89. Witch Agnesi Variant - 阿涅西箕舌线变体

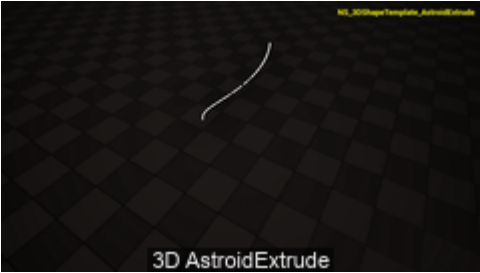


Field	Value
English	Witch Agnesi Variant
中文	阿涅西箕舌线变体
Formula (Math)	$x = t/(t^2 + 1), y = (t^2 - 1)/(t^2 + 1)$
Formula (HLSL)	<code>float2(InputT / (InputT * InputT + 1.0), (InputT * InputT - 1.0) / (InputT * InputT + 1.0))</code>
Description	Witch of Agnesi, variant.
Type / Output	2D parametric curve / float2
Input	Module.InputT (float)
Assets	NDI NDI_2D_WitchAgnesiVariant - NE NE_2DShapeTemplate_WitchAgnesiVariant - NS NS_2D_WitchAgnesiVariant - NFS NFS2D_WitchAgnesiVariant

3D Shapes / 三维形状 (27)

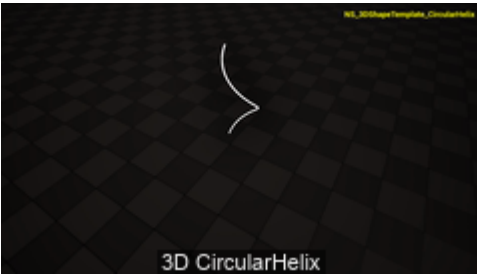
三维参数曲线 - 27

90. Astroid Extrude - 星形线拉伸体



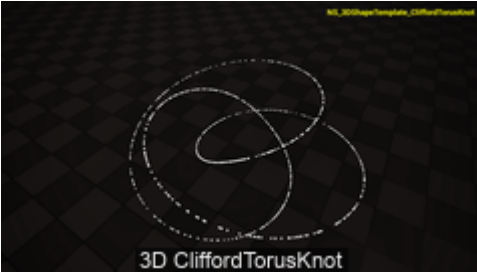
Field	Value
English	Astroid Extrude
中文	星形线拉伸体
Formula (Math)	$x = \cos^3 t, y = \sin^3 t, z = t$
Formula (HLSL)	<code>float3(pow(cos(InputT),3.0), pow(sin(InputT),3.0), InputT)</code>
Description	Astroid extruded along Z.
Type / Output	3D parametric curve / float3
Input	Module.InputT (float)
Assets	NDI NDI_3D_AstroidExtrude - NE NE_3DShapeTemplate_AstroidExtrude - NS NS_3D_AstroidExtrude - NFS NFS3D_AstroidExtrude

91. Circular Helix - 圆柱螺旋线



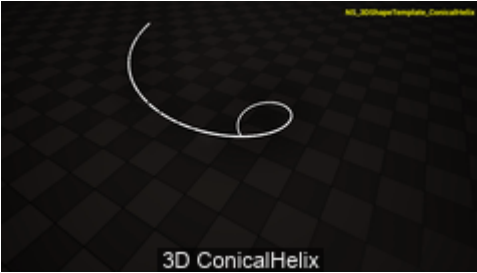
Field	Value
English	Circular Helix
中文	圆柱螺旋线
Formula (Math)	$x = \cos t, y = \sin t, z = t$
Formula (HLSL)	<code>float3(cos(InputT), sin(InputT), InputT)</code>
Description	Circular helix (constant radius).
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT (float)</code>
Assets	NDI <code>NDI_3D_CircularHelix</code> - NE <code>NE_3DShapeTemplate_CircularHelix</code> - NS <code>NS_3D_CircularHelix</code> - NFS <code>NFS3D_CircularHelix</code>

92. Clifford Torus Knot - 克利福德环面纽结



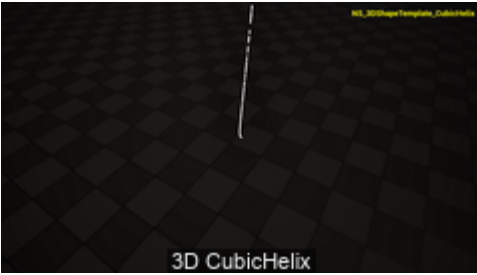
Field	Value
English	Clifford Torus Knot
中文	克利福德环面纽结
Formula (Math)	$x = (2 + \cos 2t)\cos 3t, y = (2 + \cos 2t)\sin 3t, z = \sin 2t$
Formula (HLSL)	<code>float3((2.0+cos(2.0*InputT))*cos(3.0*InputT), (2.0+cos(2.0*InputT))*sin(3.0*InputT), sin(2.0*InputT))</code>
Description	Clifford torus knot.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT (float)</code>
Assets	NDI <code>NDI_3D_CliffordTorusKnot</code> - NE <code>NE_3DShapeTemplate_CliffordTorusKnot</code> - NS <code>NS_3D_CliffordTorusKnot</code> - NFS <code>NFS3D_CliffordTorusKnot</code>

93. Conical Helix - 圆锥螺旋线



Field	Value
English	Conical Helix
中文	圆锥螺旋线
Formula (Math)	$x = t\cos t, y = t\sin t, z = t$
Formula (HLSL)	<code>float3(InputT * cos(InputT), InputT * sin(InputT), InputT)</code>
Description	Conical helix — radius grows with t.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT (float)</code>
Assets	NDI <code>NDI_3D_ConicalHelix</code> - NE <code>NE_3DShapeTemplate_ConicalHelix</code> - NS <code>NS_3D_ConicalHelix</code> - NFS <code>NFS3D_ConicalHelix</code>

94. Cubic Helix - 三次螺旋线



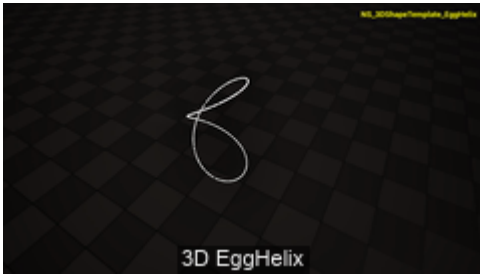
Field	Value
English	Cubic Helix
中文	三次螺旋线
Formula (Math)	$x = t, y = t^2, z = t^3$
Formula (HLSL)	<code>float3(InputT, InputT*InputT, InputT*InputT*InputT)</code>
Description	Cubic helix (t, t^2, t^3).
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT (float)</code>
Assets	NDI <code>NDI_3D_CubicHelix</code> - NE <code>NE_3DShapeTemplate_CubicHelix</code> - NS <code>NS_3D_CubicHelix</code> - NFS <code>NFS3D_CubicHelix</code>

95. Cycloidal Helix - 摆线螺旋



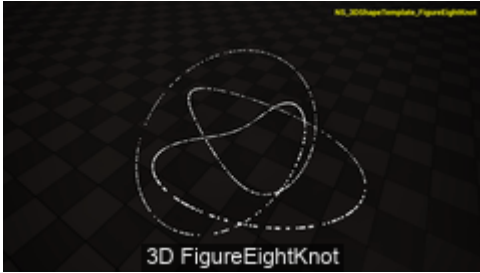
Field	Value
English	Cycloidal Helix
中文	摆线螺旋
Formula (Math)	$x = \cos t\sin t, y = \sin^2 t, z = \cos t$
Formula (HLSL)	<code>float3(cos(InputT)*sin(InputT), sin(InputT)*sin(InputT), cos(InputT))</code>
Description	Cycloidal helix.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT (float)</code>
Assets	NDI <code>NDI_3D_CycloidalHelix</code> - NE <code>NE_3DShapeTemplate_CycloidalHelix</code> - NS <code>NS_3D_CycloidalHelix</code> - NFS <code>NFS3D_CycloidalHelix</code>

96. Egg Helix - 蛋形螺旋线



Field	Value
English	Egg Helix
中文	蛋形螺旋线
Formula (Math)	$x = \cos^2 t, y = \cos t \sin t, z = \sin t$
Formula (HLSL)	<code>float3(cos(InputT)*cos(InputT), cos(InputT)*sin(InputT), sin(InputT))</code>
Description	Egg-shaped helix.
Type / Output	3D parametric curve / float3
Input	Module.InputT (float)
Assets	NDI NDI_3D_EggHelix - NE NE_3DShapeTemplate_EggHelix - NS NS_3D_EggHelix - NFS NFS3D_EggHelix

97. Figure Eight Knot - 八字结



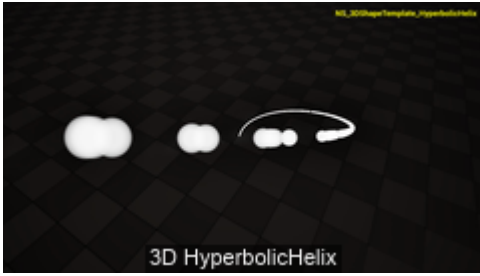
Field	Value
English	Figure Eight Knot
中文	八字结
Formula (Math)	$x = (2 + \cos 2t) \cos 3t, y = (2 + \cos 2t) \sin 3t, z = \sin 4t$
Formula (HLSL)	<code>float3((2.0+cos(2.0*InputT))*cos(3.0*InputT), (2.0+cos(2.0*InputT))*sin(3.0*InputT), sin(4.0*InputT))</code>
Description	Figure-eight knot.
Type / Output	3D parametric curve / float3
Input	Module.InputT (float)
Assets	NDI NDI_3D_FigureEightKnot - NE NE_3DShapeTemplate_FigureEightKnot - NS NS_3D_FigureEightKnot - NFS NFS3D_FigureEightKnot

98. Great Circle Sphere - 球面大圆



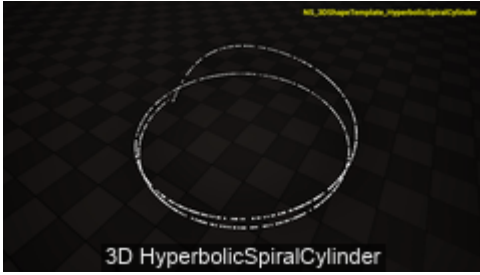
Field	Value
English	Great Circle Sphere
中文	球面大圆
Formula (Math)	$x = \cos(t/2) \cos t, y = \cos(t/2) \sin t, z = \sin(t/2)$
Formula (HLSL)	<code>float3(cos(InputT*0.5)*cos(InputT), cos(InputT*0.5)*sin(InputT), sin(InputT*0.5))</code>
Description	Great circle on a sphere.
Type / Output	3D parametric curve / float3
Input	Module.InputT (float)
Assets	NDI NDI_3D_GreatCircleSphere - NE NE_3DShapeTemplate_GreatCircleSphere - NS NS_3D_GreatCircleSphere - NFS NFS3D_GreatCircleSphere

99. Hyperbolic Helix - 双曲螺旋线



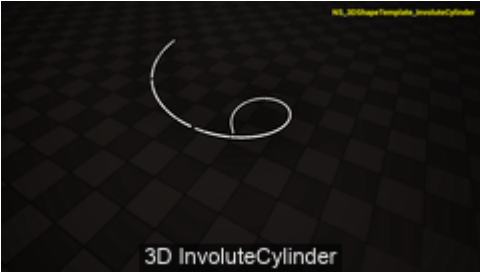
Field	Value
English	Hyperbolic Helix
中文	双曲螺旋线
Formula (Math)	$x = \cosh t \cos t, y = \cosh t \sin t, z = \sinh t$
Formula (HLSL)	<code>float3(cosh(InputT)*cos(InputT), cosh(InputT)*sin(InputT), sinh(InputT))</code>
Description	Hyperbolic helix (cosh/sinh).
Type / Output	3D parametric curve / float3
Input	Module.InputT (float)
Assets	NDI NDI_3D_HyperbolicHelix - NE NE_3DShapeTemplate_HyperbolicHelix - NS NS_3D_HyperbolicHelix - NFS NFS3D_HyperbolicHelix

100. Hyperbolic Spiral Cylinder - 双曲螺旋柱面



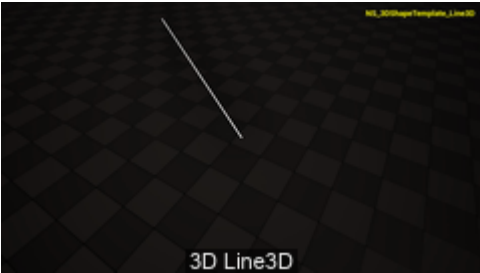
Field	Value
English	Hyperbolic Spiral Cylinder
中文	双曲螺旋柱面
Formula (Math)	$x = \cos t, y = \sin t, z = t/(t^2 + 1)$
Formula (HLSL)	<code>float3(cos(InputT), sin(InputT), InputT/(InputT*InputT+1.0))</code>
Description	Hyperbolic spiral wrapped on a cylinder.
Type / Output	3D parametric curve / float3
Input	Module.InputT (float)
Assets	NDI NDI_3D_HyperbolicSpiralCylinder - NE NE_3DShapeTemplate_HyperbolicSpiralCylinder - NS NS_3D_HyperbolicSpiralCylinder - NFS NFS3D_HyperbolicSpiralCylinder

101. Involute Cylinder - 渐开线柱面



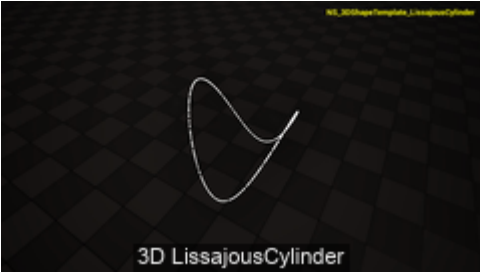
Field	Value
English	Involute Cylinder
中文	渐开线柱面
Formula (Math)	$x = \cos t + t \sin t, y = \sin t - t \cos t, z = t$
Formula (HLSL)	<code>float3(cos(InputT)+InputT*sin(InputT), sin(InputT)-InputT*cos(InputT), InputT)</code>
Description	Involute wrapped on a cylinder.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI <code>NDI_3D_InvoluteCylinder</code> - NE <code>NE_3DShapeTemplate_InvoluteCylinder</code> - NS <code>NS_3D_InvoluteCylinder</code> - NFS <code>NFS3D_InvoluteCylinder</code>

102. Line 3D - 三维直线



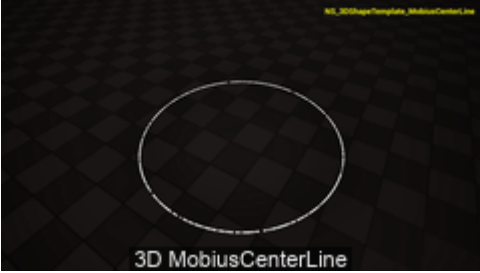
Field	Value
English	Line 3D
中文	三维直线
Formula (Math)	$x = t, y = 0.5t, z = 0.3t$
Formula (HLSL)	<code>float3(InputT, InputT*0.5, InputT*0.3)</code>
Description	Straight 3D line.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI <code>NDI_3D_Line3D</code> - NE <code>NE_3DShapeTemplate_Line3D</code> - NS <code>NS_3D_Line3D</code> - NFS <code>NFS3D_Line3D</code>

103. Lissajous Cylinder - 利萨茹柱面



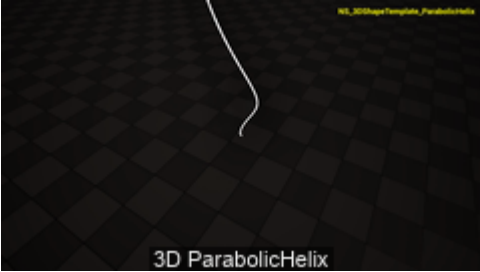
Field	Value
English	Lissajous Cylinder
中文	利萨茹柱面
Formula (Math)	$x = \cos t, y = \sin t, z = \cos 2t$
Formula (HLSL)	<code>float3(cos(InputT), sin(InputT), cos(2.0*InputT))</code>
Description	Lissajous figure wrapped on a cylinder.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI <code>NDI_3D_LissajousCylinder</code> - NE <code>NE_3DShapeTemplate_LissajousCylinder</code> - NS <code>NS_3D_LissajousCylinder</code> - NFS <code>NFS3D_LissajousCylinder</code>

104. Mobius Center Line - 莫比乌斯中心线



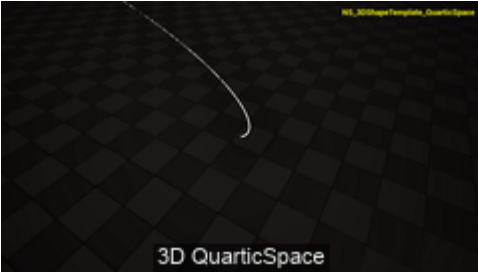
Field	Value
English	Mobius Center Line
中文	莫比乌斯中心线
Formula (Math)	$x = \cos t, y = \sin t, z = 0$
Formula (HLSL)	<code>float3(cos(InputT), sin(InputT), 0.0)</code>
Description	Centre line of a Mobius strip.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI <code>NDI_3D_MobiusCenterLine</code> - NE <code>NE_3DShapeTemplate_MobiusCenterLine</code> - NS <code>NS_3D_MobiusCenterLine</code> - NFS <code>NFS3D_MobiusCenterLine</code>

105. Parabolic Helix - 抛物螺旋线



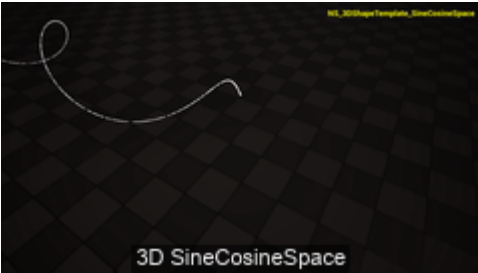
Field	Value
English	Parabolic Helix
中文	抛物螺旋线
Formula (Math)	$x = t \cos t, y = t \sin t, z = t^2$
Formula (HLSL)	<code>float3(InputT*cos(InputT), InputT*sin(InputT), InputT*InputT)</code>
Description	Parabolic helix — radius and height grow with t.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI <code>NDI_3D_ParabolicHelix</code> - NE <code>NE_3DShapeTemplate_ParabolicHelix</code> - NS <code>NS_3D_ParabolicHelix</code> - NFS <code>NFS3D_ParabolicHelix</code>

106. Quartic Space - 四次空间曲线



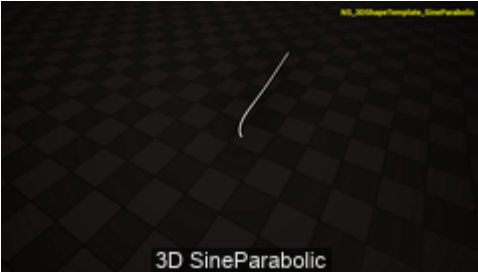
Field	Value
English	Quartic Space
中文	四次空间曲线
Formula (Math)	$x = t^4 - 2t^2, y = t^3, z = t^2$
Formula (HLSL)	<code>float3(pow(InputT,4.0)-2.0*InputT*InputT, InputT*InputT*InputT, InputT*InputT)</code>
Description	Quartic space curve.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI <code>NDI_3D_QuarticSpace</code> - NE <code>NE_3DShapeTemplate_QuarticSpace</code> - NS <code>NS_3D_QuarticSpace</code> - NFS <code>NFS3D_QuarticSpace</code>

107. Sine Cosine Space - 正弦余弦空间曲线



Field	Value
English	Sine Cosine Space
中文	正弦余弦空间曲线
Formula (Math)	$x = t, y = \sin t, z = \cos t$
Formula (HLSL)	<code>float3(InputT, sin(InputT), cos(InputT))</code>
Description	Sine/cosine space curve.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI <code>NDI_3D_SineCosineSpace</code> - NE <code>NE_3DShapeTemplate_SineCosineSpace</code> - NS <code>NS_3D_SineCosineSpace</code> - NFS <code>NFS3D_SineCosineSpace</code>

108. Sine Parabolic - 正弦抛物线



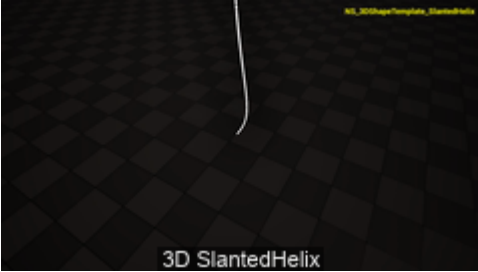
Field	Value
English	Sine Parabolic
中文	正弦抛物线
Formula (Math)	$x = t, y = t^2, z = \sin t$
Formula (HLSL)	<code>float3(InputT, InputT*InputT, sin(InputT))</code>
Description	Sine/parabolic space curve.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI <code>NDI_3D_SineParabolic</code> - NE <code>NE_3DShapeTemplate_SineParabolic</code> - NS <code>NS_3D_SineParabolic</code> - NFS <code>NFS3D_SineParabolic</code>

109. Sinusoidal Cylinder - 正弦柱面



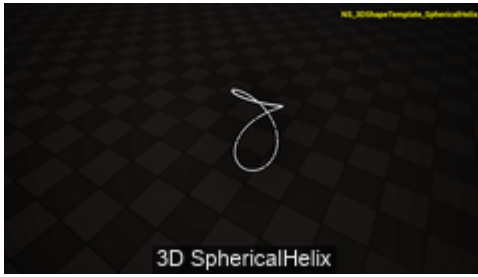
Field	Value
English	Sinusoidal Cylinder
中文	正弦柱面
Formula (Math)	$x = \cos t, y = \sin t, z = \sin t$
Formula (HLSL)	<code>float3(cos(InputT), sin(InputT), sin(InputT))</code>
Description	Sinusoidal cylinder.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI <code>NDI_3D_SinusoidalCylinder</code> - NE <code>NE_3DShapeTemplate_SinusoidalCylinder</code> - NS <code>NS_3D_SinusoidalCylinder</code> - NFS <code>NFS3D_SinusoidalCylinder</code>

110. Slanted Helix - 倾斜螺旋线



Field	Value
English	Slanted Helix
中文	倾斜螺旋线
Formula (Math)	$x = \cos t, y = \sin t, z = t^2$
Formula (HLSL)	<code>float3(cos(InputT), sin(InputT), InputT*InputT)</code>
Description	Helix with a quadratic Z rise.
Type / Output	3D parametric curve / <code>float3</code>
Input	<code>Module.InputT</code> (float)
Assets	NDI <code>NDI_3D_SlantedHelix</code> - NE <code>NE_3DShapeTemplate_SlantedHelix</code> - NS <code>NS_3D_SlantedHelix</code> - NFS <code>NFS3D_SlantedHelix</code>

111. Spherical Helix - 球面螺旋线



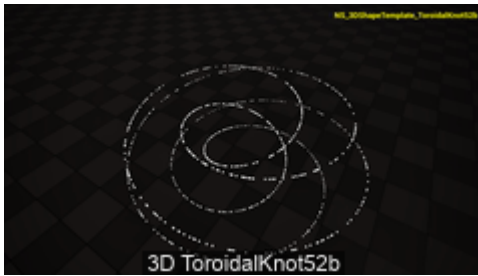
Field	Value
English	Spherical Helix
中文	球面螺旋线
Formula (Math)	$x = \cos t \sin t, y = \sin^2 t, z = \cos t$
Formula (HLSL)	<code>float3(cos(InputT)*sin(InputT), sin(InputT)*sin(InputT), cos(InputT))</code>
Description	Spherical helix.
Type / Output	3D parametric curve / float3
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_3D_SphericalHelix - NE NE_3DShapeTemplate_SphericalHelix - NS NS_3D_SphericalHelix - NFS NFS3D_SphericalHelix

112. Toroidal Knot 52 - 环面纽结 (5,2)



Field	Value
English	Toroidal Knot 52
中文	环面纽结 (5,2)
Formula (Math)	$x = \cos 3t \cos 2t, y = \cos 3t \sin 2t, z = \sin 3t$
Formula (HLSL)	<code>float3(cos(3.0*InputT)*cos(2.0*InputT), cos(3.0*InputT)*sin(2.0*InputT), sin(3.0*InputT))</code>
Description	Torus knot with p=5, q=2.
Type / Output	3D parametric curve / float3
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_3D_ToroidalKnot52 - NE NE_3DShapeTemplate_ToroidalKnot52 - NS NS_3D_ToroidalKnot52 - NFS NFS3D_ToroidalKnot52

113. Toroidal Knot 52b - 环面纽结 (5,2) 变体



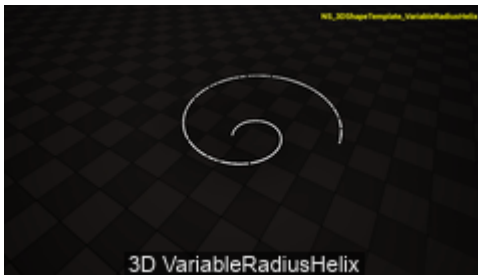
Field	Value
English	Toroidal Knot 52b
中文	环面纽结 (5,2) 变体
Formula (Math)	$x = (2 + \cos 2t) \cos 5t, y = (2 + \cos 2t) \sin 5t, z = \sin 2t$
Formula (HLSL)	<code>float3((2.0+cos(2.0*InputT))*cos(5.0*InputT), (2.0+cos(2.0*InputT))*sin(5.0*InputT), sin(2.0*InputT))</code>
Description	Torus knot (5,2), variant.
Type / Output	3D parametric curve / float3
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_3D_ToroidalKnot52b - NE NE_3DShapeTemplate_ToroidalKnot52b - NS NS_3D_ToroidalKnot52b - NFS NFS3D_ToroidalKnot52b

114. Trefoil Knot - 三叶结



Field	Value
English	Trefoil Knot
中文	三叶结
Formula (Math)	$x = \sin t + 2 \sin 2t, y = \cos t - 2 \cos 2t, z = -\sin 3t$
Formula (HLSL)	<code>float3(sin(InputT)+2.0*sin(2.0*InputT), cos(InputT)-2.0*cos(2.0*InputT), -sin(3.0*InputT))</code>
Description	Trefoil knot.
Type / Output	3D parametric curve / float3
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_3D_TrefoilKnot - NE NE_3DShapeTemplate_TrefoilKnot - NS NS_3D_TrefoilKnot - NFS NFS3D_TrefoilKnot

115. Variable Radius Helix - 变半径螺旋线



Field	Value
English	Variable Radius Helix
中文	变半径螺旋线
Formula (Math)	$x = (1 + 0.5t) \cos t, y = (1 + 0.5t) \sin t, z = 0.3t$
Formula (HLSL)	<code>float3((1.0+0.5*InputT)*cos(InputT), (1.0+0.5*InputT)*sin(InputT), InputT*0.3)</code>
Description	Helix whose radius grows with t.
Type / Output	3D parametric curve / float3
Input	<code>Module.InputT (float)</code>
Assets	NDI NDI_3D_VariableRadiusHelix - NE NE_3DShapeTemplate_VariableRadiusHelix - NS NS_3D_VariableRadiusHelix - NFS NFS3D_VariableRadiusHelix

116. Viviani - 维维亚尼曲线



Field	Value
English	Viviani
中文	维维亚尼曲线
Formula (Math)	$x = \cos^2 t, y = \cos t \sin t, z = \sin t$
Formula (HLSL)	<code>float3(cos(InputT)*cos(InputT), cos(InputT)*sin(InputT), sin(InputT))</code>
Description	Viviani curve (sphere $\cap$ cylinder).
Type / Output	3D parametric curve / float3
Input	<code>Module.InputT</code> (float)
Assets	NDI NDI_3D_Viviani - NE NE_3DShapeTemplate_Viviani - NS NS_3D_Viviani - NFS NFS3D_Viviani

3D Special (multi-parameter) / 三维特殊形状 (12)

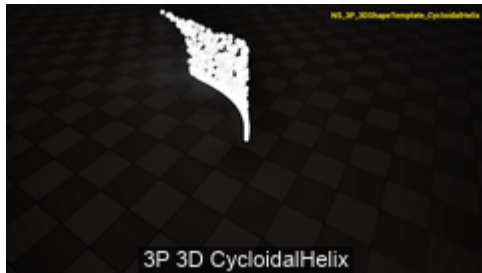
多参数三维曲线 - 12

117. Circular Helix - 圆柱螺旋线 (3参数)



Field	Value
English	Circular Helix
中文	圆柱螺旋线 (3参数)
Formula (Math)	$x = u \cdot \cos t, y = u \cdot \sin t, z = v \cdot t$
Formula (HLSL)	<code>float3(InputU*cos(InputT), InputU*sin(InputT), InputV*InputT)</code>
Description	3D circular helix — $x = r \cos(t), y = r \sin(t), z = c \cdot t$.
Type / Output	3D parametric curve (3 parameters) / float3
Input	<code>Module.InputT</code> (float, t), <code>Module.InputU</code> (float, r), <code>Module.InputV</code> (float, c)
Assets	NDI NDI_3P_3D_CircularHelix - NE NE_3P_3DShapeTemplate_CircularHelix - NS NS_3P_3DShapeTemplate_CircularHelix - NFS NFS_3P_3D_CircularHelix

118. Cycloidal Helix - 摆线螺旋 (3参数)



Field	Value
English	Cycloidal Helix
中文	摆线螺旋 (3参数)
Formula (Math)	$x = u(t - \sin t), y = u(1 - \cos t), z = v \cdot t$
Formula (HLSL)	<code>float3(InputU*(InputT-sin(InputT)), InputU*(1.0-cos(InputT)), InputV*InputT)</code>
Description	3D cycloidal helix — $x = a(t - \sin(t)), y = a(1 - \cos(t)), z = b \cdot t$.
Type / Output	3D parametric curve (3 parameters) / float3
Input	<code>Module.InputT</code> (float, t), <code>Module.InputU</code> (float, a), <code>Module.InputV</code> (float, b)
Assets	NDI NDI_3P_3D_CycloidalHelix - NE NE_3P_3DShapeTemplate_CycloidalHelix - NS NS_3P_3DShapeTemplate_CycloidalHelix - NFS NFS_3P_3D_CycloidalHelix

119. Involute Cylinder - 渐开线柱面 (3参数)



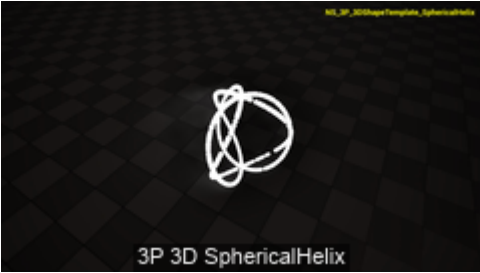
Field	Value
English	Involute Cylinder
中文	渐开线柱面 (3参数)
Formula (Math)	$x = u(\cos t + t \sin t), y = u(\sin t - t \cos t), z = v \cdot t$
Formula (HLSL)	<code>float3(InputU*(cos(InputT)+InputT*sin(InputT)), InputU*(sin(InputT)-InputT*cos(InputT)), InputV*InputT)</code>
Description	3D involute on cylinder — $x = R(\cos(t) + t \sin(t)), y = R(\sin(t) - t \cos(t)), z = c \cdot t$.
Type / Output	3D parametric curve (3 parameters) / float3
Input	<code>Module.InputT</code> (float, t), <code>Module.InputU</code> (float, R), <code>Module.InputV</code> (float, c)
Assets	NDI NDI_3P_3D_InvoluteCylinder - NE NE_3P_3DShapeTemplate_InvoluteCylinder - NS NS_3P_3DShapeTemplate_InvoluteCylinder - NFS NFS_3P_3D_InvoluteCylinder

120. Slanted Helix - 倾斜螺旋线 (3参数)



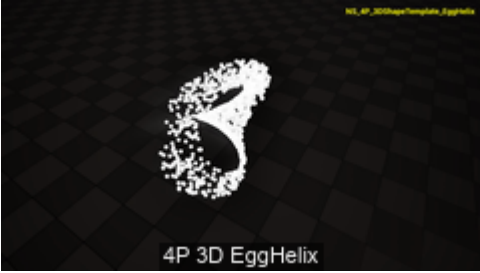
Field	Value
English	Slanted Helix
中文	倾斜螺旋线 (3参数)
Formula (Math)	$x = u \cdot \cos t, y = u \cdot \sin t, z = v \cdot t^2$
Formula (HLSL)	<code>float3(InputU*cos(InputT), InputU*sin(InputT), InputV*InputT*InputT)</code>
Description	3D slanted helix — $x = a \cos(t), y = a \sin(t), z = b \cdot t^2$.
Type / Output	3D parametric curve (3 parameters) / float3
Input	<code>Module.InputT</code> (float, t), <code>Module.InputU</code> (float, a), <code>Module.InputV</code> (float, b)
Assets	NDI NDI_3P_3D_SlantedHelix - NE NE_3P_3DShapeTemplate_SlantedHelix - NS NS_3P_3DShapeTemplate_SlantedHelix - NFS NFS_3P_3D_SlantedHelix

121. Spherical Helix - 球面螺旋线 (3参数)



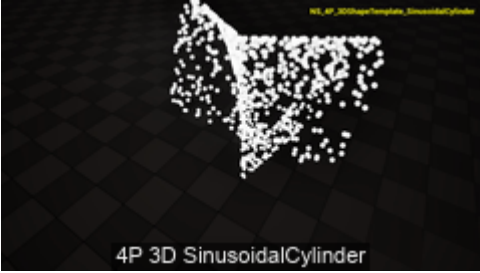
Field	Value
English	Spherical Helix
中文	球面螺旋线 (3参数)
Formula (Math)	$x = u \cdot \cos t \sin(vt), y = u \cdot \sin t \sin(vt), z = u \cdot \cos(ct)$
Formula (HLSL)	<code>float3(InputU*cos(InputT)*sin(InputV*InputT), InputU*sin(InputT)*sin(InputV*InputT), InputU*cos(InputV*InputT))</code>
Description	3D spherical helix — $x=acos(t)\sin(ct), y=asin(t)\sin(ct), z=acos(ct)$.
Type / Output	3D parametric curve (3 parameters) / float3
Input	<code>Module.InputT (float, t), Module.InputU (float, a), Module.InputV (float, c)</code>
Assets	NDI NDI_3P_3D_SphericalHelix - NE NE_3P_3DShapeTemplate_SphericalHelix - NS NS_3P_3DShapeTemplate_SphericalHelix - NFS NFS_3P_3D_SphericalHelix

122. Egg Helix - 蛋形螺旋线 (4参数)



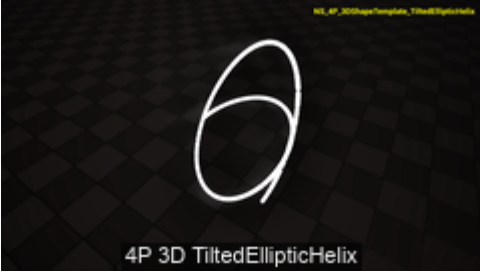
Field	Value
English	Egg Helix
中文	蛋形螺旋线 (4参数)
Formula (Math)	$x = (a + b \cdot \cos t) \cdot \cos t, y = (a + b \cdot \cos t) \cdot \sin t, z = c \cdot \sin t$
Formula (HLSL)	<code>float3((InputA+InputB*cos(InputT))*cos(InputT), (InputA+InputB*cos(InputT))*sin(InputT), InputC*sin(InputT))</code>
Description	3D egg-shaped helix — $x=(a+b\cos(t))\cos(t), y=(a+b\cos(t))\sin(t), z=c\sin(t)$.
Type / Output	3D parametric curve (4 parameters) / float3
Input	<code>Module.InputT (float, t), Module.InputA (float, a), Module.InputB (float, b), Module.InputC (float, c)</code>
Assets	NDI NDI_4P_3D_EggHelix - NE NE_4P_3DShapeTemplate_EggHelix - NS NS_4P_3DShapeTemplate_EggHelix - NFS NFS_4P_3D_EggHelix

123. Sinusoidal Cylinder - 正弦柱面 (4参数)



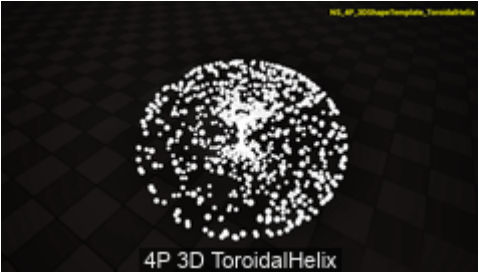
Field	Value
English	Sinusoidal Cylinder
中文	正弦柱面 (4参数)
Formula (Math)	$x = R \cdot \cos t, y = R \cdot \sin t, z = a \cdot \sin(kt)$
Formula (HLSL)	<code>float3(InputR*cos(InputT), InputR*sin(InputT), InputA*sin(InputK*InputT))</code>
Description	3D sinusoidal curve on cylinder — $x=R\cos(t), y=R\sin(t), z=Asin(kt)$.
Type / Output	3D parametric curve (4 parameters) / float3
Input	<code>Module.InputT (float, t), Module.InputR (float, R), Module.InputA (float, A), Module.InputK (float, k)</code>
Assets	NDI NDI_4P_3D_SinusoidalCylinder - NE NE_4P_3DShapeTemplate_SinusoidalCylinder - NS NS_4P_3DShapeTemplate_SinusoidalCylinder - NFS NFS_4P_3D_SinusoidalCylinder

124. Tilted Elliptic Helix - 倾斜椭圆螺旋线 (4参数)



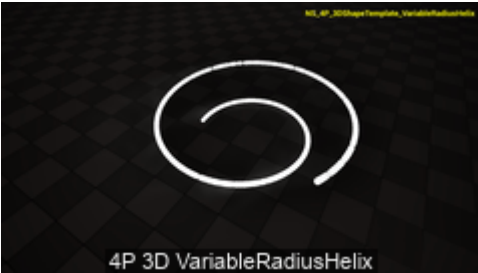
Field	Value
English	Tilted Elliptic Helix
中文	倾斜椭圆螺旋线 (4参数)
Formula (Math)	$x = a \cdot \cos t, y = b \cdot \sin t, z = c \cdot t \cdot \sin t$
Formula (HLSL)	<code>float3(InputA*cos(InputT), InputB*sin(InputT), InputC*InputT*sin(InputT))</code>
Description	3D tilted elliptic helix — $x=acos(t), y=bsin(t), z=ctsint(t)$.
Type / Output	3D parametric curve (4 parameters) / float3
Input	<code>Module.InputT (float, t), Module.InputA (float, a), Module.InputB (float, b), Module.InputC (float, c)</code>
Assets	NDI NDI_4P_3D_TiltedEllipticHelix - NE NE_4P_3DShapeTemplate_TiltedEllipticHelix - NS NS_4P_3DShapeTemplate_TiltedEllipticHelix - NFS NFS_4P_3D_TiltedEllipticHelix

125. Toroidal Helix - 环形螺旋线 (4参数)



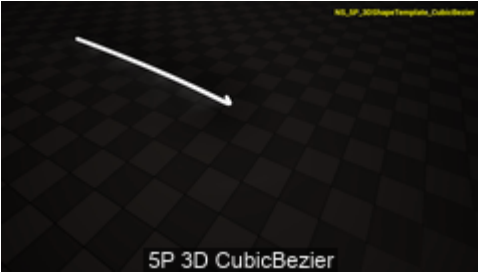
Field	Value
English	Toroidal Helix
中文	环形螺旋线 (4参数)
Formula (Math)	$x = (R + p \cdot \cos(nt)) \cdot \cos t, y = (R + p \cdot \cos(nt)) \cdot \sin t, z = p \cdot \sin(nt)$
Formula (HLSL)	<code>float3((InputR+InputRho*cos(InputN*InputT))*cos(InputT), (InputR+InputRho*cos(InputN*InputT))*sin(InputT), InputRho*sin(InputN*InputT))</code>
Description	3D toroidal helix — $x=(R+rcos(nt))cos(t), y=(R+rcos(nt))sin(t), z=rsin(nt)$.
Type / Output	3D parametric curve (4 parameters) / <code>float3</code>
Input	<code>Module.InputT</code> (float, t), <code>Module.InputR</code> (float, R), <code>Module.InputRho</code> (float, r), <code>Module.InputN</code> (float, n)
Assets	NDI NDI_4P_3D_ToroidalHelix - NE NE_4P_3DShapeTemplate_ToroidalHelix - NS NS_4P_3DShapeTemplate_ToroidalHelix - NFS NFS_4P_3D_ToroidalHelix

126. Variable Radius Helix - 变半径螺旋线 (4参数)



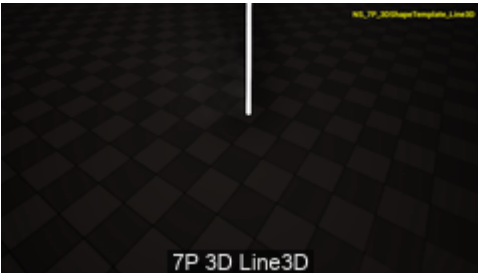
Field	Value
English	Variable Radius Helix
中文	变半径螺旋线 (4参数)
Formula (Math)	$x = (a + b \cdot t) \cdot \cos t, y = (a + b \cdot t) \cdot \sin t, z = c \cdot t$
Formula (HLSL)	<code>float3((InputA+InputB*InputT)*cos(InputT), (InputA+InputB*InputT)*sin(InputT), InputC*InputT)</code>
Description	3D variable-radius helix — $x=(a+bt)cos(t), y=(a+bt)sin(t), z=c \cdot t$.
Type / Output	3D parametric curve (4 parameters) / <code>float3</code>
Input	<code>Module.InputT</code> (float, t), <code>Module.InputA</code> (float, a), <code>Module.InputB</code> (float, b), <code>Module.InputC</code> (float, c)
Assets	NDI NDI_4P_3D_VariableRadiusHelix - NE NE_4P_3DShapeTemplate_VariableRadiusHelix - NS NS_4P_3DShapeTemplate_VariableRadiusHelix - NFS NFS_4P_3D_VariableRadiusHelix

127. Cubic Bezier - 三次贝塞尔曲线 (5参数)



Field	Value
English	Cubic Bezier
中文	三次贝塞尔曲线 (5参数)
Formula (Math)	$P(t) = (1 - t)^3 P_0 + 3(1 - t)^2 t P_1 + 3(1 - t) t^2 P_2 + t^3 P_3$
Formula (HLSL)	<code>pow(1.0-InputT,3.0)*InputP0 + 3.0*pow(1.0-InputT,2.0)*InputT*InputP1 + 3.0*(1.0-InputT)*InputT*InputT*InputP2 + InputT*InputT*InputT*InputP3</code>
Description	3D cubic Bezier curve — $P(t)=(1-t)^3P0+3(1-t)^2tP1+3(1-t)t^2P2+t^3P3$.
Type / Output	3D parametric curve (5 parameters) / <code>float3</code>
Input	<code>Module.InputT</code> (float, t), <code>Module.InputP0</code> ... <code>Module.InputP3</code> (float3, control points (vec3))
Assets	NDI NDI_5P_3D_CubicBezier - NE NE_5P_3DShapeTemplate_CubicBezier - NS NS_5P_3DShapeTemplate_CubicBezier - NFS NFS_5P_3D_CubicBezier

128. Line 3D - 三维直线 (7参数)



Field	Value
English	Line 3D
中文	三维直线 (7参数)
Formula (Math)	$x = x_0 + a \cdot t, y = y_0 + b \cdot t, z = z_0 + c \cdot t$
Formula (HLSL)	<code>float3(InputX0+InputA*InputT, InputY0+InputB*InputT, InputZ0+InputC*InputT)</code>
Description	3D line — $x=x0+at, y=y0+bt, z=z0+ct$.
Type / Output	3D parametric curve (7 parameters) / <code>float3</code>
Input	<code>Module.InputT</code> (float, t), <code>Module.InputX0</code> (float, x0), <code>Module.InputY0</code> (float, y0), <code>Module.InputZ0</code> (float, z0), <code>Module.InputA</code> (float, a), <code>Module.InputB</code> (float, b), <code>Module.InputC</code> (float, c)
Assets	NDI NDI_7P_3D_Line3D - NE NE_7P_3DShapeTemplate_Line3D - NS NS_7P_3DShapeTemplate_Line3D - NFS NFS_7P_3D_Line3D